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IDENTIFICATION

Product Code: Maindec-08-D3RA-D
Product Name: DECTREX 1
TC01 Random Exerciser
Maindec 851
Date: January 9, 1967
Maintainer: Diagnostic Group
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PDP-8
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1. ABSTRACT

DECTREX 1 is a DECTape Random Exerciser for the TC01 DECTape control and any configuration of one to eight TU55 DECTape transports. Drive selection, tape direction, number of blocks, sequence of operation and patterns generated are by random selection. The DECTape functions exercised are search, read data and write data in normal and continuous modes, read all in continuous mode, and move.

Also included are a short series of processor tests that are executed while waiting for interrupts and during data breaks while searching, reading, and writing from DECTape.

2. REQUIREMENTS

2.1 Equipment

PDP-8 (standard)

TC01 DECTape Control

One to eight TU55 DECTape Transports

One standard PDP-8 DECTape for each drive (2702₈ 129-word blocks)

2.2 Storage

2.2.1 Program Storage - The program occupies most of memory from address 0000 to 5000. In addition the JMS test uses addresses 6000 to 6200 for JMS storage.

2.2.2 Buffer Areas - The program utilizes three 129-word buffer areas as follows:

<u>Addresses</u>	<u>Contents</u>
6774 to 7174	Random data and output buffer
7175 to 7375	Read buffer 1
7376 to 7576	Read buffer 2

(Read buffers are used for WRITE DATA C MODE)

2.3 Preliminary Programs

All parts of the TC01 Basic Exerciser should run before attempting to run DECTREX 1.

3. LOADING PROCEDURE

3.1 Method

Procedures of normal binary loading from paper tape should be followed.

4. STARTING PROCEDURE

4.1 Control Switch Settings

When initially starting the program, SWITCH REGISTER bits 0 to 7 are used for drive selection. Each bit is a master bit for a drive. When the switch is a1 the drive is selected; when a0 the drive is not selected. Switch 0 is the master bit for drive 8, switch 1 for drive 1, etc. Any configuration of switches is considered valid except all 0s.

4.2 Starting Address

The starting address for DECTREX 1 is 0200.

4.3 Program and/or Operator Action

Load DECTREX 1 into memory.
Dial the desired drive number(s) on each TU55 to be tested.
Put each TU55 ON LINE, WRITE ENABLED with a standard PDP-8 DECTape installed.
Set the SWITCH REGISTER to 0200.
Press LOAD ADDRESS.
Set the SWITCH REGISTER to select drives per paragraph 4.1.
Press START.
The processor halts at address 0207.
Set all SWITCH REGISTER bits to 0 or as desired per paragraph 5.1.
Press CONTINUE.

5. OPERATING PROCEDURE

5.1 Operational Switch Settings

SW0	UP	Delete error typeouts and halts.
SW1	UP	Delete error halts.
SW2	UP	Type first four data compare errors in each block.
SW2	DOWN	Type all data compare errors.

SW11	DOWN	Only hit end zone once for turnaround for blocks 0000 and 2701.
SW11	UP	Hit end zone twice before turnaround for blocks 0000 and 2701.

6. ERRORS

All DECTape hardware malfunctions detected by the program result in an error typeout and an error halt (see paragraph 5.1). The halt does not occur until all errors pertaining to the block and operation have been typed. (A read-data parity error and data compare error could occur in the same block. In this case halt would not occur until after the COMPARE ERROR typeout.)

The first three lines of every typeout indicate the DECTape drive, operation direction and mode, and the block being operated on or to be found as an end result of search.

6.1 Error Typeouts

6.1.1 Search Error Typeouts - The search error typeouts contain the following information:

Drive number.
 Search direction and mode.
 Block wanted and direction.
 The block number put into memory by the TC01.
 The last block number found, if more than two blocks have been found.
 The number of blocks found since the last start-up or turnaround.
 DECTape status B.

Examine the typeout in the following order:

- a. Examine the direction of search and the direction of the block wanted. If they are different, the error occurred before turnaround. If the directions indicated are the same, the error occurred after turnaround.
- b. Examine the STAT B typeout. If it is 0001, indicating a normal interrupt (DECTape flag only), it can probably be ignored (6000 would indicate a mark track error). If STAT B is anything other than 0001, it caused the error typeout.
- c. Examine the number of block numbers read (BLOCKS READ). There are three conditions to note: 0000, indicating no block numbers read since start-up or turnaround; 0001, indicating one block number was read since start-up or turnaround; 0002 or greater, indicating that more than one block number was found.

d. Examine LAST BLOCK. This line is included only if two or more normal block numbers have been read since start-up or turnaround. This line is pertinent only if STAT B equals 0001.

e. Examine BLOCK FOUND. This line holds the contents of the memory location that block numbers are read into and is pertinent only if BLOCKS READ is not equal to 0000, or the STAT B typeout indicates that the DECtape flag is set (bit 11 a 1).

If STAT B is 0001 and BLOCKS READ is 0002 or greater, BLOCK FOUND compared against LAST BLOCK indicates two block numbers read that are not numerically consecutive.

If STAT B is not 0001 and BLOCKS READ is not 0000, BLOCK FOUND indicates the block where the status error occurred.

If STAT B is 0001, BLOCKS READ is 0001, and the direction of search is the same direction as the block wanted, the error was TURN AROUND.

The typeout C MODE indicates continuous mode. BLOCKS READ in a SEARCH C MODE typeout always indicates two blocks read. LAST BLOCK indicates the last block found in normal mode.

6.1.2 Write Data Error Typeouts - Write data typeouts contain the following information:

Drive number.
Direction and C mode if continuous mode block is being written.
DECtape status B.
Contents of the word count register (address 7754).

6.1.3 Read Data Typeouts -

6.1.3.1 Read Data Status Error Typeouts - These read data typeouts include the following information:

Drive number
Direction and mode
Block being read
DECtape status B if STAT B typed = 0001 see WC (word count register).
Contents of WC if it does not equal 0000.
(Note: If WC does not equal 0, no data compare is made.)

6.1.3.2 Read Data Compare Error Typeouts - These typeouts include the drive number, direction, mode, and block read. Each pair of data words is separated by a blank line of paper. The first octal number is the data generated or regenerated by the program. The second number is data read from tape, and the third line is the memory address of the data read.

6.1.4 Read All Error Typeouts -

6.1.4.1 Read All Status Errors - Read All Status Error typeouts include drive number, read all direction and mode, block number, and DECTape status B.

6.1.4.2 Read All CHECKSUM ERROR - This typeout indicates an error in parity generation (especially if not followed by a read-all compare error) and includes drive number, read-all direction and mode, block number, and the following:

a. REVERSE CHECKSUM as read from tape. If read-all direction is backward, the complement obverse of this line is the checksum going forward.

b. DATA CHECKSUM CALCULATED as generated by the 6-bit XOR of the data by the program. Since the data is written to be bidirectional, this line indicates the same sum for either direction.

c. CHECKSUM as read from tape is in the upper six bits of the word typed (bits 0 to 5). Again, if the read-all direction is backward, the complement obverse of these six bits would be the REV CKSUM if read forward.

d. LPB CALCULATED is the sum of the first three lines, and is generated by the same process as the TC01 parity generation circuitry. The LPB should equal 77_8 after the process is complete. The 0 bits in the lower six bits (bits 6 to 11) of this typeout are the error(s) that caused the typeout.

6.1.4.3 Read All Compare Error - These typeouts follow the same format as read-data compare errors. If a read-all compare error follows a checksum error, ignore the checksum error.

6.1.5 Program Interrupt Errors - The program also detects three classes of program interrupt errors.

a. Program interrupt and the DECTape IOT 771 did not skip. PI NO DECTAPE SKIP

b. No program interrupt for 45 seconds if a MOVE tape or for 5 seconds if any other DECTape function and DECTape IOT 771 did not skip at the end of that period.
NO PI NO DECTAPE SKIP

c. No program interrupt for 45 seconds if a MOVE tape or for 5 seconds if any other DECTape function and the DECTape IOT 771 did skip.
NO PI DECTAPE SKIP

Following one of these typeouts, the program forces another error typeout from the routine that called the wait for interrupt. The second typeout indicates the exact operation that the DECtape was doing when the PI error occurred.

6.1.6 Processor Errors - Any processor errors detected by the program result in an error halt only. Consult the program listing to determine the cause of the halt. The following table contains processor error halts and a description of the error:

<u>Address</u>	
3211	ISZ failed. Address 3342 should = 0.
3215	ISZ failed. Address 3341 should = 0001.
3226	ROTATE 1. Link should = 1.
3232	ROTATE 1. Data failure in AC address 3342 is data tested.
3242	ROTATE 2. Link should = 0.
3246	ROTATE 2. Data failure in AC. Address 3342 is data tested.
3272	TAD failure. Address 3347 plus address 3343 should = 3344.
3331	JMS failure. Address 3343 points to an address that does not equal itself + 1 after executing a JMS.

6.1.7 Examples of Error Typeouts -

6.1.7.1 Example 1 Search Errors -

DRIVE 8

SEARCH FWD

2677 BLOCK WANTED FWD

2700 BLOCK FOUND

0001 BLOCKS READ

0001 STAT B

(Block searched)

(Block found)

(One block number received)

(Normal interrupt)

This typeout indicates that the DECtape drive did not turn around and come up to speed in time.

DRIVE 8

SEARCH BKWD
0000 BLOCK WANTED FWD
1033 BLOCK FOUND
0000 BLOCKS READ
6000 STAT B

(Error was a mark-track error if STAT B is not normal interrupt and BLOCKS READ = 0000. Ignore second block number)

DRIVE 2

SEARCH BKWD
0077 BLOCK WANTED BKWD
0105 BLOCK FOUND
0000 BLOCKS READ
5000 STAT B

This timeout occurs if an end zone interrupt is received in error.

DRIVE 6

SEARCH FWD
2701 BLOCK WANTED FWD
0000 BLOCK FOUND
7700 LAST BLOCK
0002 BLOCKS READ
0001 STAT B

This timeout indicates that two block numbers were found in search and that they were not sequential.

6.1.7.2 Example 2 Read Data Status Error -

DRIVE 4

READ DATA FORWARD
0265 BLOCK
4200 STAT B

(Block being read)
(Parity error)
(If any data errors, a second timeout follows)

DRIVE 3

READ DATA BACKWARD
0011 BLOCK
4400 STAT B
7577 WC

(Block being read)
(Select error)
(WC was not = 0. The program makes no data comparison in this case.)

DRIVE 5

READ DATA FWD C MODE
0227 BLOCK
0001 STAT B
7577 WC

(Note: Normal STAT B)
(Error was WC did not go to 0000 before DTF was set)

6.1.7.3 Example 3 Write Data Status Error -

DRIVE 1

WRITE DATA FWD
0001 BLOCK
6000 STAT B
7743 WC

A mark-track error was received while doing a write data on block 1.

DRIVE 1

WRITE DATA BKWD C MODE
0040 BLOCK
0001 STAT B
7400 WC

(Note: Normal STAT B)
(WC indicates that WC overflow had not occurred when DTF was set).

6.1.7.4 Example 4 Read all Checksum Error -

DRIVE 6

READ ALL BACKWARD C MODE
0175 BLOCK
CHECKSUM ERROR

0033 REV CKSUM
0022 DATA CKSUM CALCULATED
6700 CKSUM
0076 CALCULATED CKSUM

(The complements of the REVCK, DATA and CKSUM did not XOR to 77)
(as read from tape)
(XOR of the data's complement)
(as read from tape)
(XOR of the complement of the above. Any 0 bits are in error. Calculated CKSUM should = 0077.)

NOTE: In this case, the REV CKSUM would be the CKSUM going forward. Realize also, that it would be complement obverse (44) going forward.

Since the DATA FORMAT written by DECTREX is bidirectional, the DATA CHECKSUM CALCULATED would be the same read in either direction.

Also, the CKSUM going forward would be the REV CKSUM; the complement obverse of 67 would be 01.

Note that the calculated checksum would be 67, if the block had been read forward.

6.1.7.5 Example 5 Read Compare Error - Read-data and read-all-data compare errors follow the same format.

DRIVE 4

READ DATA FORWARD
DATA ERROR
0265 BLOCK

(or READ ALL C MODE)

4632 COR
4432 INC
7237 ADRS INC

(Pattern word generated)
(Pattern word read)
(Memory address of the incorrect data)

2315 COR
2115 INC
7240 ADRS INC

6.2 Error Recovery

After an error typeout, the processor halts. Press CONTINUE to recover. For all errors, the program attempts to repeat the same or a similar operation. If a second error is encountered for the same operation, the typeout and HALT again occurs. However, pressing CONTINUE the second time causes the drive in error to rewind, and the random selections for that drive begin again from block 0. If a second error is not encountered, the program completes the operation and then continues its normal random selections.

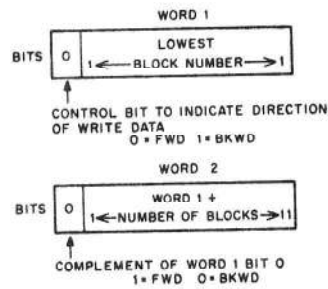
For the read errors (read data and read all) the direction of the second read operation is opposite the direction that the error was originally detected in. If a second error is detected, a typeout and halt again occur. Pressing CONTINUE causes the drive to be rewound to the end zone. If no error occurs during the second pass, the block is read a third time. The third read is in the same direction as the first read, or the direction the error was originally detected in. Again, if any errors are detected, a typeout and halt occur. In either case, the end result is that the drive is rewound and random selections start from block 0, so that the same read error sequence is not generated again.

7. RESTRICTIONS (None)

8. MISCELLANEOUS

8.1 Data Format

The data blocks written by DECTREX 1 are formatted to be bidirectional. Whether written forward or backward, they may be read in either direction on the TC01 with the resultant data looking the same. The first four words of the block contain pertinent information about the block. The first two words of each block are formatted as follows:



WORD 1 indicates the lowest numbered block in the series with bit 0 indicating the written direction of the series of blocks.

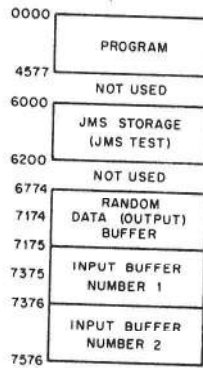
WORD 2 is 1 greater than the highest numbered block in the series bit 0 in word 2 is the complement of bit 0 in WORD 1.

Ignoring bits 0 of the two words, WORD 2 - WORD 1 = number of blocks in the series.

Word 3 and word 4 of the block are random numbers used to generate the rest of the data. Word 5 of the block is word 3 rotated right one position. Word 6 is word 4 rotated right one position. This process is repeated until word 64 of the block is word 62 rotated right. The lower six bits of word 65 of the block contain the number of 1 bits in words 3 and 4. The upper six bits of word 65 (middle word of 129) are the complement obverse of the lower six bits. Word 66 is the complement obverse of word 64. Word 67 is the complement obverse of word 63, etc., until the last word of the block is the complement obverse of the first word of the block.

WORD 1	4036	Blocks 36, 37, 40, and
WORD 2	0042	41 were written backward
WORD 3	3252	with these two words
WORD 4	7734	as key to data pattern.
WORD 5	1525	WORD 3 RAR
WORD 6	3756	WORD 4 RAR
	etc.	WORD 5 RAR WORD 6 RAR
WORD 65	0617	Number of bits in words 3 and 4 in bits 6 to 11. Complement obverse of same in bits 0 to 5.
WORD 126	3400	Complement obverse of word 4
WORD 127	1204	Complement obverse of word 3
WORD 128	5377	Complement obverse of word 2
WORD 129	1473	Complement obverse of word 1

8.2 Core Map



9. PROGRAM DESCRIPTION

9.1 Discussion

DECTREX 1 is a random exerciser for the TC01 DECTape Control and any configuration of one to eight TU55 DECTape Drives. Drive, direction of operation, number of blocks, and data patterns are by random selection. First the program randomly selects a drive, then a number between 1 and 32 decimal for the number of blocks, and next the direction. There is one possibility in four that the direction will be backward. The number of blocks is added to or subtracted from the last block position of the drive selected. If the block generated has not been written, a write operation is initiated. If the last block table indicates that the block selected has already been written, a read operation is initiated. If READ is selected, the program then generates a random number between 0 and 7. If the number is 0, the block is read in read-all continuous mode; if 4, read-data normal mode; if any other number, read data continuous mode and two blocks are read. If WRITE is selected and the number of blocks is a multiple of 3 (3, 6, 9, etc.), the blocks are written in continuous mode. Otherwise, the blocks are written in normal mode.

The processor tests, that are run while waiting for interrupts do not guarantee that the processor operates correctly; but, other than EAE operations, should allow a reasonable amount of confidence in the processor.



WRITE1 1400
 WRTDTY 0121
 WRT1A 1455
 WSTERR 1526
 WTCNTR 3125
 WTKON1 3123
 WTKON2 3124
 XORSAY 2565
 XORSUM 2531

/TC01 RANDOM EXERCISER
 /PAGE 0 CONSTANTS AND TEMP STOR.
 RANBFR=6774
 BUFFRS=RANBFR+201
 IOT=6000

/GO TO PROCESS INTERRUPT

*1

0001 5402 JMP I .+1
 0002 3034 IRECD

*40

0040	0000	RECORD,	0	/SEARCH ID
0041	0000	BLKFND,	0	/BLOCK FROM TC01
0042	0041	IDCON,	.-1	/FOR SRCH CA
0043	7754	WCLOC,	7754	/TO STORE WC
0044	7755	CALOC,	7755	/TO STORE CA
0045	0000	CDRIVE,	0	/CURRENT DRIVE NUMBER
0046	0000	JNFUNC,	0	/SAME FOR TC01 COMMAND
0047	0000	MSBITS,	0	/DRIVE SELECT SWITCHES
0050	0000	COMBIT,	0	
0051	0000	POSITN,	0	/TO GET DRIVE POSITION
0052	0000	DIRECT,	0	/TO GET DRIVE DIRECTION
0053	0000	-STBLK,	0	/TO GET LAST BLOCK WRITTEN
0054	0000	-STDRV,	0	/-LAST DRV SELECTED

/SUBROUTINE ADDRESSES

0055	3600	SRCHIT,	SEARCH
0056	3400	REWIND,	REPOSI
0057	3437	NEWDRV,	CHNGDR
0060	1000	SELRAV,	RANSEL
0061	0600	GENPAT,	PATGEN
0062	0735	RANGEN,	GENRAN
0063	0400	READ1,	RDDATA
0064	0547	REGENP,	PREGEN
0065	1225	DATACO,	CODATA
0066	3000	WAITI,	WATINT
0067	4321	ERRSTP,	ERSTP
0070	1200	DATAMV,	MVDATA
0071	0040	RECRDK,	RECORD
0072	7777	COCNTR,	7777
0073	6774	RBUFST,	RANBFR
0074	6773	RBFLC,	RANBFR-1
0075	7174	BF1LOC,	BUFFRS-1
0076	0000	JIRFLG,	0
0077	0000	VUMBLK,	0
0100	7174	RBUFND,	RANBFR+200
0101	4441	WRTDTY,	TYWDAT
0102	6775	RBFW2,	RANBFR+1
0103	7376	BF2LOC,	BUFFRS+201
0104	0420	READ2,	REREAD
0105	0470	READ2A,	REREDA
0106	4272	TYPCON,	TYCONT
0107	4200	TYPTX,	TYTEXT

/-1 NO DATA ERRORS

/MASK AND COMPARE CONSTANTS

0110	7760	<7760,	7760
0111	0037	<0037,	37
0112	7770	<7770,	7770
0113	0200	<0200,	200
0114	0040	<0040,	40
0115	0614	<0614,	614
0116	0030	<0030,	30
0117	7577	<7577,	7577
0120	0007	<0007,	7
0121	4000	<4000,	4000
0122	0700	<0700,	700
0123	0070	<0070,	70
0124	7000	<7000,	7000
0125	0003	<0003,	3
0126	0214	<0214,	214
0127	7700	<7700,	7700
0130	4511	DRIVTY,	TYDRV
0131	1000	<1000,	1000
E2BIT=K1000			
0132	0010	<0010,	10
0133	0020	<0020,	20
0134	0077	<0077,	77
0135	0100	<0100,	100
0136	0240	<0240,	240
0137	0400	<0400,	400
0140	0604	<0604,	604
0141	0016	<16,	16
0142	0260	<260,	260
0143	3777	<3777,	3777
0144	4423	READTY,	TYRDAT
0145	4411	SRCHTY,	TYSRCH
0146	5075	<5075,	5075
0147	5076	<5076,	5076
0150	0050	<0050,	50
0151	0000	PASFLG,	0
0152	0002	<0002,	2
0153	2701	<2701,	2701
0154	6000	<6000,	6000

/CONSTANTS FOR SEARCH ROUTINE

0155	4000	SRCHER,	SRHERR
0156	0000	TAPONT,	0
0157	0000	BLKFLG,	0
0160	0000	PREBLK,	0

/JMS RETURN FOR PROCESSOR TEST

0161	3321	RETUJM,	JMRETU
0162	0000	FRSWAT,	0

/TC01 DECTAPE
 /RANDOM DATA RANDOM SEQUENCE EXERCISER
 /TESTS CONCURRENT OPERATION OF ANY CONFIGURATION OF 1 TO 8 DECTAPE
 /DRIVES.
 /MASTER BIT SELECTION BIT 0=DRV8 BIT1=DRV1 TO BIT7=DRV7

*200

0200	7604	RANDEX,	CLA OSR	
0201	0110	AND	K7760	
0202	7440	SZA		/SELECT ANY AT ALL
0203	5206	JMP	,+3	/YES
0204	7402	HLT		/DIT NOT SELECT ANY DRIVES
0205	5200	JMP	RANDEX	
0206	3047	DCA	MSBITS	
0207	7402	HLT		/WAIT CLR SELECTION
0210	7040	CMA		
0211	3152	DCA	FRSWAT	
0212	6774	IOT	774	/CLEAR STAT B
0213	4456	JMS	I REWIND	/PUT ALL DRIVES IN END ZONE
0214	3453	DCA	I LSTBLK	/CLR NUMBER OF BLOCKS
0215	4457	JMS	I NEWDRV	/CLR ALL
0216	5214	JMP	, -2	/NO
0217	7201	IAC	CLA	
0220	3054	DCA	LSTDRV	
/MEAT OF THE PROGRAM				
/RANDOM SELECTION OF OPERATIONS				
/FIRST SELECT DRIVE				
0221	4460	MOFPRO,	JMS I SELRAN	/RANDOMLY SELECT A DRIVE
0222	7240	CLA	CMA	
0223	3151	DCA	PASFLG	
0224	1045	TAD	CDRIVE	
0225	1054	TAD	LSTDRV	
0226	7650	SNA	CLA	
0227	5325	JMP	SAMDRV	
0230	1045	TAD	CDRIVE	
0231	7041	CMA	IAC	
0232	3054	DCA	LSTDRV	
0233	7040	CMA		
0234	3350	DCA	RSOFLG	
/SELECT 1 TO 32 BLOCKS FOR OPERATION				
0235	4462	MOFPR1,	JMS I RANGEN	/GET RAN NUMBER
0236	0111	AND	K0037	
0237	7040	CMA		/MAKE -1 TO -32 DECI
0240	3077	DCA	NUMBLK	

```

/SELECT DIRECTION TO GO
0241 4462 JMS I RANGEN /GET RANDOM NUMBER
0242 0125 AND K0003
0243 7650 SNA CLA
0244 7040 CMA /SET BACKWARDS
0245 3076 DCA DIRFLG /DIRECTION FLAG
0246 1453 TAD I LSTBLK
0247 7650 SNA CLA /DRIVE BEEN WRITTEN ON
0250 5724 JMP I WRITES /NO, WRITE
0251 7100 CLL
0252 1076 TAD DIRFLG

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/GENERATE BLOCK SELECT POS + OR-NUMBER
0253 7640 SZA CLA
0254 7120 STL
0255 1077 TAD NUMBLK
0256 7420 SNL
0257 7041 CMA IAC
0260 1451 TAD I POSITN
0261 3040 DCA RECORD
0262 1040 TAD RECORD
0263 7710 SPA CLA /-RECORD
0264 3040 DCA RECORD /MAKE =0
/HAS BLOCK SELECTED BEEN WRITTEN
/IF NOT WRITE OPERATION SELECTED
0265 1040 TAD RECORD
0266 7040 CMA
0267 1453 TAD I LSTBLK
0270 7710 SPA CLA /BLOCK BEEN WRITTEN
0271 5724 JMP I WRITES /NO
0272 4462 JMS I RANGEN
0273 0120 AND K0007
0274 7450 SNA
0275 5703 JMP I .+6
0276 0125 AND K0003
0277 7650 SNA CLA
0300 5304 JMP .+4
0301 5702 JMP I .+1
0302 2000 RDCMOD
0303 2400 RALLTS
0304 4463 RDTAB, JMS I READ1
0305 5504 JMP I READ2 /STATUS ERROR RETURN

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```

/BLOCK HAS BEEN READ WITHOUT PAR ERR
0326 1113 TAD K0200
0327 6764 IOT 764 /STOP TAPE
0312 1040 TAD RECORD
0311 3451 DCA I POSITN /NEW POSITION
0312 1076 TAD DIRFLG
0313 3452 DCA I DIRECT /DIRECTION READ

/MOVE FIRST FOUR WORDS TO REGENERATE
0314 4464 JMS I REGENP /REGERATE DATA PATTERN
0315 4465 JMS I DATACO
0316 6774 RANBFR
0317 7175 BUFFRS
0320 0201 201
0321 2072 ISZ COCNTR
0322 5505 JMP I READ2A
0323 5221 JMP MOFPRO /SELECT NEW DRIVE
0324 1400 WRITES, WRITE1

/DRIVE SELECTED SAME AS LAST TIME TEST FOR SERIES
0325 4462 SAMDRV, JMS I RANGEN
0326 7510 SPA /READ THIS STRING
0327 5236 JMP MOFPR1+1 /FIND NEW BLOCK
0330 2350 ISZ RSQFLG
0331 5236 JMP MOFPR1+1
0332 0125 AND K0003
0333 7650 SNA CLA /FWD
0334 7040 CMA /NO GO BACKWARD
0335 3076 DCA DIRFLG
0336 1076 TAD DIRFLG
0337 7041 CMA IAC
0340 1073 TAD RBUFST /FWD GET
0341 3040 DCA RECORD /FIRST BLOCK
0342 1076 TAD DIRFLG /BKWD GETS
0343 1440 TAD I RECORD /LST BLOCK+1-1
0344 0143 AND K3777
0345 3040 DCA RECORD
0346 5747 JMP I .+1
0347 1600 RDSEQ
0350 0000 RSQFLG, 0

```

/READ DATA SUBROUTINE 1 BLOCK
/DIRFLG=7777 IS BACKWARDS=0 IS FWD

*400

0400	5200	RDDATA,	JMP .	
0401	7200	CLA		
0402	4455	JMS I SRCHIT		
0403	1116	TAD K0030	/SEARCH TO READ DATA	
0404	6764	IOT 764		
0405	1117	TAD K7577	/129 WORDS	
0406	3443	DCA I WCLOC		
0407	1075	TAD BF1LOC	/INTO FIRST BUFFER	
0410	3444	DCA I CALOC		
0411	4466	JMS I WAITI		
0412	0001	1		
0413	5600	JMP I RDDATA	/STATUS ERR(READ EXIT)	
0414	1443	TAD I WCLOC		
0415	7650	SNA CLA		
0416	2200	ISZ RDDATA		
0417	5600	JMP I RDDATA	/NORMAL READ EXIT	

/TYPE STATUS ERROR ON READ 1 BLOCK
/TEST FOR DATA ERRORS IF FULL READ

0420	7240	REREAD,	CLA CMA	
0421	3101	DCA PASFLG		
0422	4407	JMS I ERRSTR	/STOP TAPE	
0423	4544	JMS I READTY		
0424	1271	TAD RECRDK		
0425	4506	JMS I TYPCON	/TYPE BLOCK NUMBER	
0426	4507	JMS I TYPTX		
0427	2242	0042		
0430	5457	5457		
0431	4353	4353		
0432	7700	7700		
0433	6772	10T 772		
0434	3010	DCA 10		
0435	1132	TAD K0010		
0436	4506	JMS I TYPCON	/TYPE STATUS B	
0437	4507	JMS I TYPTX		
0440	2053	0063		
0441	6441	6441		
0442	6400	6400		
0443	4200	4200		
0444	7777	7777		
0445	7700	7700		
0446	1443	TAD I WCLOC		
0447	7650	SNA CLA	/READ 129 WORD	
0450	5260	JMP .+10	/YES	
0451	1043	TAD WCLOC		
0452	4506	JMS I TYPCON	/SHORT BUFFER	
0453	4507	JMS I TYPTX	/TYPEOUT	
0454	2057	0067		
0455	1643	1643		
0456	1677	1677		
0457	5270	JMP REREDA		
0460	4464	JMS I REGENP	/REGENERATE PATTERN	
0461	1444	TAD I CALOC		
0462	1343	TAD K7600		
0463	3266	DCA .+3		
0464	4455	JMS I DATACO	/COMPARE FOR TYPEOUTS	
0465	6774	RANBFR		
0466	7175	BUFFRS		
0467	2201	0201		

```

0470 7624 REREDA, LAS
0471 0154 AND K6200
0472 7650 SNA CLA
0473 7402 HLT
0474 2151 ISZ PASFLG
0475 5325 JMP REWDRV
0476 1076 TAD DIRFLG
0477 7040 CMA
0500 3076 DCA DIRFLG
0501 4463 JMS I READ1 /READ OPPOSITE DIRECTION
0502 5222 JMP REREAD+2 /ERROR, AGAIN, TRY OVER
0503 4464 JMS I REGENP /REGEN PATTERN
0504 4465 JMS I DATACO /COMPARE
0505 6774 RANBFR
0506 7175 BUFFRS
0507 0201 0201

0510 2072 ISZ COCNTR
0511 5270 JMP REREDA

0512 1076 TAD DIRFLG
0513 7040 CMA
0514 3076 DCA DIRFLG
0515 4463 JMS I READ1

0516 5222 JMP REREAD+2 /ANOTHER STATUS ERROR
0517 4465 JMS I DATACO
0520 6774 RANBFR
0521 7175 BUFFRS
0522 0201 0201
0523 2072 ISZ COCNTR
0524 5270 JMP REREDA

0525 1140 REWDRV, TAD K0604
0526 1046 TAD UNFUNC
0527 6766 IOT 766
0530 4466 JMS I WAITI
0531 5000 5000
0532 5344 JMP K7600+1
0533 3453 DCA I LSTBLK
0534 3451 DCA I POSITN
0535 7040 CMA
0536 3452 DCA I DIRECT
0537 7001 IAC
0540 3054 DCA LSTDRV
0541 5742 JMP I .+1
0542 0221 MOFPRO
0543 7600 K7600, 7600
0544 4746 JMS I .+2
0545 5325 JMP REWDRV
0546 3542 MOVER+2

```

/MOVE FIRST 4 WORDS OF
/BLOCK READ AND REGENERATE
/COMPARE PATTERN

0547	5347	PREGEN,	JMP ,
0552	1274	TAD	RBFL0C
0551	3210	DCA	10
0552	1372	TAD	GETRED+2
0553	3017	DCA	17
0554	1444	TAD	I CALOC
0555	3370	DCA	GETRED
0556	1770	TAD	I GETRED
0557	4771	JMS	I GETRED+1
0560	3410	DCA	I 10
0561	7040	CMA	
0562	1370	TAD	GETRED
0563	3370	DCA	GETRED
0564	2017	ISZ	17
0565	5356	JMP	,-7
0566	4461	JMS	I GENPAT
0567	5747	JMP	I PREGEN
0570	0000	GETRED,	0
0571	0703	MCOMOB	
0572	7774	7774	

/GENERATE RANDOM DATA PATTERN
 /FIRST FOUR WORDS OF PATTERN ARE IN
 /RANDOM BUFFER WORDS 1 TO 4

```

*600
0600 5200 PATGEN, JMP ,
0601 1353 TAD MIN4
0602 3275 DCA GENDEX /TO COUNT FIRST 4 WORDS
0603 1073 TAD RBUFST
0604 3276 DCA GENDEX+1 /TO STORE FROM START
0605 1100 TAD RBUFND
0606 3277 DCA GENDEX+2 /TO STORE FROM END
0607 1301 TAD RPATO+1 /TO GET
0610 3300 DCA RPATO /RANDJMS
0611 1676 TAD I GENDEX+1
0612 3301 DCA RPATO+1
0613 1301 TAD RPATO+1
0614 4303 JMS MCOMOB /MAKE COMPLIMENT OBVERSE
0615 3677 DCA I GENDEX+2
0616 2276 ISZ GENDEX+1 /INC ADDRESS
0617 7240 CLA CMA
0620 1277 TAD GENDEX+2 /DECREMENT OTHER
0621 2275 ISZ GENDEX /DONE FIRST 4
0622 5206 JMP PATGEN+6 /NO

0623 3277 DCA GENDEX+2
0624 1354 TAD MIN60 /TO COUNT 60 WORDS
0625 3275 DCA GENDEX
0626 1300 GNPATR, TAD RPATO
0627 7110 CLL RAR /FORM NEXT
0630 7430 SZL /WORD OF
0631 1121 TAD K4000 /RANDOM PATTERN
0632 3302 DCA RPATO+2

0633 1302 TAD RPATO+2
0634 3676 DCA I GENDEX+1
0635 1302 TAD RPATO+2
0636 4303 JMS MCOMOB
0637 3677 DCA I GENDEX+2
0640 1301 TAD RPATO+1 /MOVE WORDS
0641 3300 DCA RPATO /FOR NEXT PASS
0642 1302 TAD RPATO+2
0643 3301 DCA RPATO+1
0644 2276 ISZ GENDEX+1 /INCREMENT LOWER
0645 7240 CLA CMA
0646 1277 TAD GENDEX+2 /DECREMENT UPPER ADDRESS
0647 3277 DCA GENDEX+2
0650 2275 ISZ GENDEX /DONE ALL
0651 5226 JMP GNPATR /NO

```

0652	3302	DCA RPATO+2	/CLR FOR COUNTING BITS
0653	1300	TAD RPATO	
0654	7110	CLL RAR	
0655	7430	SZL	/BIT=1
0656	2302	ISZ RPATO+2	/YES, COUNT
0657	7440	SZA	/DONE, FIRST WORD
0660	5254	JMP , -4	/NO
0661	1301	TAD RPATO+1	/GET 2ND
0662	7110	CLL RAR	
0663	7430	SZL	/BIT=1
0664	2302	ISZ RPATO+2	/YES
0665	7440	SZA	/DONE 2ND
0666	5262	JMP , -4	/NO
0667	1302	TAD RPATO+2	
0670	4303	JMS MCOMOB	/MAKE OBVERSE
0671	0127	AND K7700	
0672	1302	TAD RPATO+2	/MAKE WORD 65
0673	3676	DCA I GENDEX+1	
0674	5600	JMP I PATGEN	/EXIT
0675	0000	GENDEX, 0	/TO COUNT WORDS
0676	0000	0	/TO STORE LOWER
0677	0000	0	/TO STORE UPPER
0700	0000	RPATO, 0	/PATTERN STORAGE
0701	0000	0	
0702	0000	0	/TO COUNT BITS

/MAKE COMPLIMENT OBVERSE OF AC

```

0703 5303 MCOMOB,      JMP .
0704 3355      DCA COMSTR      /SAVE ORIGINAL
0705 1355      TAD COMSTR
0706 0122      AND K0702
0707 7110      CLL RAR
0710 7012      RTR
0711 3356      DCA COMSTR+1
0712 1355      TAD COMSTR

0713 0123      AND K0070
0714 7006      RTL
0715 7004      RAL
0716 1356      TAD COMSTR+1
0717 3356      DCA COMSTR+1
0720 1355      TAD COMSTR
0721 0120      AND K0007
0722 7012      RTR
0723 7012      RTR
0724 1356      TAD COMSTR+1
0725 3356      DCA COMSTR+1
0726 1355      TAD COMSTR
0727 0124      AND K7000
0730 7006      RTL
0731 7006      RTL
0732 1356      TAD COMSTR+1
0733 7040      CMA
0734 5703      JMP I MCOMOB

```

/RANDOM NUMBER GENERATOR

```

0735 5335 GENRAN,      JMP .
0736 7200      CLA
0737 1352      TAD RANVAR
0740 7104      CLL RAL
0741 7430      SZL
0742 7001      IAC
0743 3352      DCA RANVAR
0744 1352      TAD RANVAR
0745 1351      TAD RANNO
0746 3351      DCA RANNO
0747 1351      TAD RANNO
0750 5735      JMP I GENRAN      /EXIT AC=RANNO

```

```

0751 2634 RANNO,      2634
0752 4263 RANVAR,     4263
0753 7774 MIN4,       7774
0754 7724 MIN60,      7724
0755 0000 COMSTR,     0

```

/RANDOMLY SELECT A DRIVE
 /STAYS IN THIS ROUTINE UNTIL A DRIVE
 /IS FOUND AND DIRECTION AND POSITION
 /POINTERS ARE GENERATED

```

*1000
1000 5200 RANSEL,      JMP .
1001 4246      JMS SELGEN      /GET RANDOM NUMBER
1002 0120      AND K0007      /SELECT DRIVE
1003 3045      DCA CDRIVE
1004 1045      TAD CDRIVE
1005 7110      CLL RAR
1006 7012      RTR
1007 7010      RAR

1010 3046      DCA UNFUNC      /POSITION DRIVE NUMBER
1011 1045      TAD CDRIVE
1012 7040      CMA
1013 3010      DCA 10          /MAKE NEG FOR COUNT
1014 1121      TAD K4000
1015 3011      DCA 11
1016 2010      ISZ 10          /BIT IN DRIVE POSITION
1017 7410      SKP

1020 5225      JMP EXIST      /BIT IS IN POSITION
1021 1011      TAD 11
1022 7110      CLL RAR      /MOVE UNIT BIT
1023 3011      DCA 11
1024 5216      JMP , -6
1025 1011      EXIST, TAD 11      /GET UNIT BIT
1026 0047      AND MSBITS      /MASK WITH DRIVES SELECT
1027 7650      SNA CLA      /DOES DRIVE EXIST

1030 5201      JMP RANSEL+1      /NO, TRY AGAIN
1031 1045      TAD CDRIVE
1032 1243      TAD POSTBL
1033 3051      DCA POSITN      /FORM POSITION POINTER
1034 1045      TAD CDRIVE
1035 1244      TAD DIRTBL
1036 3052      DCA DIRECT      /FORM DIRECTION POINTER
1037 1045      TAD CDRIVE
1040 1245      TAD LSTTBL      /GENERATE LAST BLOCK
1041 3053      DCA LSTBLK      /WRITTEN POINTER
1042 5600      JMP I RANSEL

```

```

1043 3502 POSTBL,    PNTRS+1      /TO GET DRIVE POSITION
1044 3513 DIRTBL,    PNTRS+12     /DIRECTION
1045 3524 LSTTBL,    PNTRS+23     /LAST BLOCK WRITTEN
1046 5246 SELGEN,    JMP .
1047 1261      TAD SELVAR
1050 7104      CLL RAL
1051 7430      SZL
1052 7001      IAC
1053 3261      DCA SELVAR
1054 1261      TAD SELVAR
1055 1262      TAD SELVAR+1
1056 3262      DCA SELVAR+1
1057 1262      TAD SELVAR+1
1060 5646      JMP I SELGEN
1061 4263 SELVAR,    4263
1062 2634      2634

```

```

/MOVE DATA SUBROUTINE
/FROM ADDRESS=JMS+1 TO IS JMS+2
/NUMBER OF WORDS IS JMS+3

```

```

*1200
1200 5200 MVDATA,    JMP .
1201 7200      CLA
1202 1600      TAD I MVDATA
1203 2200      ISZ MVDATA
1204 3222      DCA MOVDEX
1205 1600      TAD I MVDATA
1206 2200      ISZ MVDATA
1207 3223      DCA MOVDEX+1

1210 1600      TAD I MVDATA
1211 2200      ISZ MVDATA
1212 3224      DCA MOVDEX+2
1213 1622      TAD I MOVDEX
1214 3623      DCA I MOVDEX+1
1215 2222      ISZ MOVDEX
1216 2223      ISZ MOVDEX+1
1217 2224      ISZ MOVDEX+2
1220 5213      JMP , -5
1221 5600      JMP I MVDATA

1222 0000 MOVDEX,    0
1223 0000      0
1224 0000      0

```

/COMPARE DATA SUBROUTINE
/JMS+1=ADDRESS JMS+2=ADDRESS JMS+3=WC

```

1225 5225 CODATA,      JMP .
1226 7200      CLA
1227 1625      TAD I CODATA      /GET ADDRESS 1 CORRECT
1230 2225      ISZ CODATA
1231 3255      DCA CODEX
1232 1625      TAD I CODATA      /GET ADDRESS 2 UNKNOWN
1233 3256      DCA CODEX+1
1234 2225      ISZ CODATA

1235 1625      TAD I CODATA      /GET LENGTH
1236 7041      CMA IAC
1237 3257      DCA CODEX+2
1240 2225      ISZ CODATA
1241 7040      CMA
1242 3072      DCA COCNTR      /SET NO ERROR FLAG
1243 1655      COLOOP,      TAD I CODEX
1244 7041      CMA IAC
1245 1656      TAD I CODEX+1

1246 7640      SZA CLA      /WORDS =
1247 5261      JMP COERRO      /NO, TYPE OUT
1250 2255      ISZ CODEX      /INCREMENT ADDRESS
1251 2256      ISZ CODEX+1
1252 2257      ISZ CODEX+2      /DONE ALL
1253 5243      JMP COLOOP      /NO
1254 5625      JMP I CODATA      /EXIT

1255 0000      CODEX,      0      /KNOWN DATA ADDRESS
1256 0000      0      /UNKNOWN DATA ADDRESS
1257 0000      0      /LENGTH
1260 1256      .#2

```

/DATA ERROR PRINTOUT

```

1261 2072      COERRO,      ISZ COCNTR      /FIRST DATA ERROR
1262 5313      JMP COERR1      /NOT FIRST
1263 4467      JMS I ERRSTP
1264 6761      IOT 761
1265 0132      AND K0010
1266 7650      SNA CLA
1267 5273      JMP .+4
1270 4672      JMS I .+2
1271 5274      JMP .+3
1272 4432      TYRALL
1273 4544      JMS I READTY      /FIRST ERROR
1274 4507      JMS I TYPTX      /TYPE HEADER

```

1275	7777	7777	
1276	4441	4441	/TYPE
1277	6441	6441	/CDATA ERROR)
1300	0045	0045	
1301	6262	6262	
1302	5762	5762	
1303	7700	7700	
1304	1071	TAD RECRDK	/TYPE THE OCTAL
1305	4506	JMS I TYPCON	/BLOCK NUMBER
1306	4507	JMS I TYPTX	
1307	0042	0042	
1310	5457	5457	/TYPE (BLOCK)
1311	4353	4353	
1312	7700	7700	
1313	7604	COERR1, CLA OSR	
1314	0131	AND K1000	
1315	7650	SNA CLA /ONLY TYPE4	
1316	5323	JMP ,+5 /TYPE ALL ERRORS	
1317	1072	TAD COCNTR	
1320	1354	TAD K7774	
1321	7700	SMA CLA /DONE 4 DATA TYPEOUTS	
1322	5250	JMP COLOOP+5	/YES, DELETE REST
1323	4507	JMS I TYPTX	
1324	7777	7777	/BLANK LINE BETWEEN
1325	7700	7700	/EACH PAIR
1326	1255	TAD CODEX	
1327	4506	JMS I TYPCON	/TYPE CORRECT IN OCTAL
1330	4507	JMS I TYPTX	
1331	0043	0043	/TYPE (COR)
1332	5762	5762	
1333	7700	7700	
1334	1256	TAD CODEX+1	
1335	4506	JMS I TYPCON	/TYPE INCOR OCTAL
1336	4507	JMS I TYPTX	/TYPE (INC)
1337	0051	0051	
1340	5643	5643	
1341	7700	7700	
1342	1260	TAD CODEX+3	
1343	4506	JMS I TYPCON	
1344	4507	JMS I TYPTX	
1345	0041	41	
1346	4444	4444	
1347	6263	6263	
1350	0051	51	
1351	5643	5643	
1352	7700	7700	
1353	5250	JMP COLOOP+5	
1354	7774	<7774, 7774	

/INITIATE WRITE OPERATIONS
 /GENERATE PATTERN WORDS
 /AND BLOCK NUMBERS

```

*1400
1400 1274 WRITE1, TAD RBFLOC
1401 3010 DCA 10
1402 1453 TAD I LSTBLK
1403 3410 DCA I 10 /FIRST BLOCK WRITTEN
1404 1077 TAD NUMBLK
1405 7041 CMA IAC
1406 1453 TAD I LSTBLK /LST BLOCK+1
1407 3040 DCA RECORD
1410 1040 TAD RECORD
1411 1146 TAD K5075
1412 7700 SMA CLA
1413 5317 JMP REWCK
1414 1040 TAD RECORD

1415 3410 DCA I 10
1416 4462 JMS I RANGEN
1417 3410 DCA I 10 /FIRST RANDOM WORD
1420 4462 JMS I RANGEN
1421 3410 DCA I 10 /2ND RANDOM WORD
1422 4462 JMS I RANGEN
1423 0125 AND K0003
1424 7650 SNA CLA
1425 7040 CMA
1426 3076 DCA DIRFLG
1427 1076 TAD DIRFLG
1430 1073 TAD RBUFST
1431 3010 DCA 10

1432 1010 TAD 10
1433 3011 DCA 11 /MAKE FIRST WORD
1434 1121 TAD K4000 /INDICATE BACKWARD
1435 1410 TAD I 10 /OR 2ND WORD
1436 3411 DCA I 11 INDICATE FORWARD
1437 4461 JMS I GENPAT /GENERATE 129 WORD PATTERN
1440 1077 TAD NUMBLK
1441 3370 DCA SAVNUM
1442 1077 SORT1R, TAD NUMBLK
1443 7040 CMA

1444 3040 DCA RECORD
1445 1076 TAD DIRFLG
1446 7640 SZA CLA /BACKWARD IS
1447 1040 TAD RECORD /LAST BLOCK+NUM BLOCKS
1450 1453 TAD I LSTBLK
1451 3240 DCA RECORD /TO FIND FIRST
/BLOCK TO BE WRITTEN

```

```

/TEST FOR WRITE DATA C MODE
/IF NUMBER OF BLOCKS=INC OF 3
1452 5653      JMP I .+1
1453 2200      WDCMOD
1454 4455      JMS I SRCHIT

1455 1150      WRT1A,      TAD K0050
1456 6764      IOT 764      /SRCH TO WRITE DATA
1457 1117      TAD K7577
1460 3443      DCA I WCLOC
1461 1074      TAD RBFLOC
1462 3444      DCA I CALOC
1463 4466      JMS I WAITI

1464 0001      1
1465 5326      JMP WSTERR
1466 1443      TAD I WCLOC
1467 7640      SZA CLA
1470 5326      JMP WSTERR
1471 1076      TAD DIRFLG      /0 OR -1
1472 7100      CLL
1473 1040      TAD RECORD
1474 7510      SPA
1475 7320      CLA STL
1476 3040      DCA RECORD
1477 7420      SNL      /RECORD-1
1500 2040      ISZ RECORD      /NO +1

1501 2077      ISZ NUMBLK      /DONE ALL BLOCKS
1502 5256      JMP WRT1A+1      /WRITE DATA 1 MORE
1503 1040      *DINC,      TAD RECORD      /NEW POSITION
1504 3451      DCA I POSITN
1505 1076      TAD DIRFLG

1506 3452      DCA I DIRECT      /DIRECTION FLAG
1507 1502      TAD I RBFWD2
1510 0143      AND K3777
1511 3453      DCA I LSTBLK
1512 1113      TAD K0200
1513 6764      IOT 764      /STOP TAPE
1514 5715      JMP I .+1
1515 0221      MOFPRO      /RANDOM SELECT AGAIN

1516 0525      REWDRV
1517 4462      REWCK,      JMS I RANGEN      /GET RANDOM NUMBER
1520 7710      SPA CLA      /+ OR -
1521 5716      JMP I REWCK-1
1522 3040      DCA RECORD      /+READ BLOCK 0
1523 3076      DCA DIRFLG      /FORWARD
1524 5725      JMP I .+1
1525 0304      RDTAB

```

1526	4467	WSTERR,	JMS I ERRSTP
1527	4501	JMS I WRTDTY	
1530	1071	TAD RECRDK	
1531	4506	JMS I TYPCON	
1532	4527	JMS I TYPTX	
1533	7242	42	
1534	5457	5457	
1535	4353	4353	
1536	7700	7700	
1537	1370	TAD SAVNUM	
1540	3077	DCA NUMBLK	
1541	6772	IOT 772	
1542	3372	DCA SAVNUM+2	
1543	1371	TAD SAVNUM+1	
1544	4506	JMS I TYPCON	
1545	4507	JMS I TYPTX	
1546	0063	0063	
1547	6441	6441	
1550	6400	6400	
1551	4277	4277	
1552	1043	TAD WCLOC	
1553	4506	JMS I TYPCON	
1554	4507	JMS I TYPTX	
1555	0067	67	
1556	1643	1643	
1557	1677	1677	
1560	7604	LAS	
1561	0154	AND K6000	
1562	7650	SNA CLA	
1563	7402	HLT	
1564	2151	ISZ PASFLG	
1565	5767	JMP I .+2	
1566	5242	JMP CORT1R	
1567	0525	REWDRV	
1570	0000	SAVNUM,	0
1571	1572	.+1	

/2ND ERROR
/REWIND DRIVE

/READ ENTIRE SEQUENCE OF
/BLOCKS AS ORIGINALLY WRITTEN

```

*1600
RDSEQ,      TAD I RBUFS      /GET LOWEST BLOCK
1600  1473      AND K3777
1601  0143      DCA RECORD
1602  3040      TAD I RBFWD2      /GET LST BLOCK +1
1603  1502      AND K3777
1604  0143      DCA NUMBLK      /MAKE NUMBER OF BLOCKS
1605  3077      TAD RECORD
1606  1040      CMA
1607  7040      TAD NUMBLK
1610  1077      DCA NUMBLK
1611  3077      JMS I RANGEN
1612  4462      CLL
1613  7100      SPA CLA /READ FWD
1614  7710      CMA      /NO BKWD
1615  7040      SZA
1616  7440      STL      /L=1 IF BKWD
1617  7120      DCA DIRFLG
1620  3076      TAD RECORD
1621  1040      SZL
1622  7430      TAD NUMBLK
1623  1077      DCA RECORD
1624  3040      TAD NUMBLK      /MAKE NUMBER OF BLKS-
1625  1077      CMA
1626  7040

1627  3077      DCA NUMBLK      /MAKE-
1630  4463      JMS I READ1      /READ FIRST BLOCK
1631  5504      JMP I READ2      /STATUS ERROR ON READ
1632  2077

```

		RDBUF1,	ISZ NUMBLK	/READ ALL
1633	5236	JMP ,+3	/NO,	
1634	1113	TAD K0200	/IF ALL BLOCKS READ	
1635	5242	JMP ,+5	/STOP TAPE	
1636	1103	TAD BF2LOC		
1637	3444	DCA I CALOC	/ADDRS IS BUFFER 2	
1640	1117	TAD K7577	/129 WORDS	
1641	3443	DCA I WCLOC		
1642	6764	IOT 764	/RESET ENABLES OR STOP TAPE	
1643	4465	JMS I DATACO	/COMPARE	
1644	6774	RANBFR /BLOCK	GENERATED	
1645	7175	BUFFERS	/AGAINST BLOCK READ	
1646	0201	0201		
1647	2072	ISZ COCNTR	/ANY DATA ERRORS	
1650	5505	JMP I READ2A	/YES, READ OTHER DIRECTION	
1651	1077	TAD NUMBLK		
1652	7650	SNA CLA	/DONE COMPLETE SERIES	
1653	5325	JMP NDOFRD	/YES	
1654	1076	TAD DIRFLG		
1655	7450	SNA		
1656	7001	IAC		
1657	1040	TAD RECORD	/LAST BLOCK +OR-1	
1660	3040	DCA RECORD		
1661	4466	JMS I WAITI		
1662	0001	1		
1663	5504	JMP I READ2	/STATUS ERROR, REREAD	
1664	1443	TAD I WCLOC		
1665	7640	SZA CLA		
1666	5504	JMP I READ2		
1667	2077	ISZ NUMBLK	/DONE ALL READS	
1670	5273	JMP ,+3		
1671	1113	TAD K0200	/DONE ALL	
1672	5277	JMP ,+5	/STOP TAPE	
1673	1075	TAD BF1LOC		
1674	3444	DCA I CALOC	/ADDRS IS FIRST BUFFER	
1675	1117	TAD K7577	/129 WORDS	
1676	3443	DCA I WCLOC		
1677	6764	IOT 764	/RESET ENABLES OR STOP TAPE	
1700	4465	JMS I DATACO	/COMPARE	
1701	6774	RANBFR	/GENERATED	
1702	7377	BUFFERS+202	/AGAINST READ	
1703	0201	0201		
1704	2072	ISZ COCNTR	/ANY COMPARE ERRORS	
1705	5505	JMP I READ2A	/YES, READ OTHER DIRECTION	
1706	1077	TAD NUMBLK		
1707	7650	SNA CLA	/READ AND COMPARED ALL	
1710	5325	JMP NDOFRD	/YES	
1711	1076	TAD DIRFLG		

```

1712 7450 SNA /LAST BLOCK + OR-
1713 7001 IAC /1
1714 1040 TAD RECORD
1715 3040 DCA RECORD
1716 4466 JMS I WAITI /WAIT TO FINISH READ
1717 0001 1
1720 5504 JMP I READ2 /STATUS ERROR, REREAD
1721 1443 TAD I WCLOC
1722 7640 SZA CLA
1723 5504 JMP I READ2
1724 5232 JMP RDBUF1 /COMPARE FIRST BUFFER

```

```

1725 1040 VDOFRD, TAD RECORD
1726 3451 DCA I POSITN /NEW POSITION
1727 1076 TAD DIRFLG
1730 3452 DCA I DIRECT /INDICATE DIRECTION
1731 5732 JMP I .+1
1732 0221 MOFPRO

```

PAUSE

```

/TC01 DECTREX 1 - TAPE 2
/READ DATA CONTINUOUS MODE
/TWO BLOCKS AND COMPARE DATA READ

```

*2000

```

2000 1076 RDCMOD, TAD DIRFLG
2001 7640 SZA CLA /FORWARDS
2002 5212 JMP ,+10 /NO
2003 1040 TAD RECORD
2004 7040 CMA
2005 1453 TAD I LSTBLK /BLOCK+1 WRITTEN
2006 7440 SZA
2007 5215 JMP ,+6
2010 5611 JMP I .+1 /BLOCK NOT WRITTEN
2011 0304 RDTAB /READ ONLY 1
2012 1040 TAD RECORD
2013 7650 SNA CLA /BLOCK 0 BACKWARDS
2014 5611 JMP I .-3 /READ ONLY 1

2015 4455 JMS I SRCHIT /FIND FIRST BLOCK
2016 1313 TAD K0130 /READ DATA
2017 6764 IOT 764 /CONTINUOUS MODE
2020 1117 TAD K7577
2021 1117 TAD K7577
2022 3443 DCA I WCLOC
2023 1075 TAD BF1LOC
2024 3444 DCA I CALOC
2025 4466 JMS I WAITI
2026 0001 1
2027 5275 JMP RDCERR /SEE WHICH BLOCK IN ERROR
2030 1443 TAD I WCLOC
2031 7640 SZA CLA
2032 5504 JMP I READ2
2033 1113 TAD K0200
2034 6764 IOT 764 /STOP TAPE

```

2040	1040	TAD RECORD	
2041	3040	DCA RECORD	
2042	4464	JMS I REGENP	/VERIFY LAST BLOCK
2043	4465	JMS I DATACO	/FIRST
2044	6774	RANBFR	
2045	7376	BUFFRS+201	
2046	0201	0201	
2047	2072	ISZ COCNTR	/ANY DATA ERRORS
2050	5505	JMP I READ2A	/YES REREAD
2051	7040		

2052	1103	CMA	
2053	3444	TAD BF2LOC	
		DCA I CALOC	
2054	4464	JMS I REGENP	
2055	1076	TAD DIRFLG	
2056	7450	SNA	
2057	7001	IAC	
2060	7041	CMA IAC	
2061	1040	TAD RECORD	
2062	3040	DCA RECORD	/VERIFY FIRST
2063	4465	JMS I DATACO	/BLOCK READ
2064	6774	RANBFR	
2065	7175	BUFFRS	
2066	0201	0201	
2067	2072	ISZ COCNTR	/ANY DATA ERRORS
2070	5505	JMP I READ2A	/YES, REREAD
2071	1040	TAD RECORD	
2072	3451	DCA I POSITN	
2073	5674	JMP I ,+1	
2074	0221	MOFPRO	
2075	1443	RDCERR, TAD I WCLOC	
2076	7450	SNA	/ERROR IN 2ND BLOCK
2077	5305	JMP ,+6	/YES COUNT BLOCK
2100	7041	CMA IAC	
2101	1117	TAD K7577	/WC-129
2102	3443	DCA I WCLOC	
2103	5704	JMP I ,+1	/TYPE STATUS ERROR
2104	0420	REREAD	
2105	1076	TAD DIRFLG	
2106	7450	SNA	/BACKWARDS-1
2107	7001	IAC	/FORWARD+1
2110	1040	TAD RECORD	
2111	3040	DCA RECORD	
2112	5704	JMP I ,-6	/TYPE STATUS ERROR
2113	0130	<0130,	130

/WRITE DATA CONTINUOUS MODE
/IF NUMBER OF BLOCKS IS AN INCREMENT OF 3

*2200

```

2200 3277 WDCMOD,      DCA TRECTR
2201 1077      TAD NUMBLK
2202 1125      TAD K0003
2203 7450      SNA              /AN INC OF 3
2204 5212      JMP SWCMOD      /YES WRITE C MODE
2205 7500      SMA              /GONE PAST 0
2206 5611      JMP I .+3        /NOT INC OF 3
2207 2277      ISZ TRECTR
2210 5202      JMP WDCMOD+2
2211 1454      WRT1A-1

2212 1277 SWCMOD,      TAD TRECTR      /MAKE NUMBER
2213 7040      CMA              /OF GROUPS OF 3
2214 3277      DCA TRECTR      /2'S COMPLIMENT
2215 4470      JMS I DATAMV      /MAKE PATTERN
2216 6774      RANBFR          /3 BUFFERS LONG
2217 7175      BUFFRS
2220 7376      7376
2221 4455      JMS I SRCHIT      /FIND FIRST BLOCK
2222 1301      TAD K0150        /WRITE DATA CONTINUOUSLY
2223 6764 SWCMDL,      IOT 764
2224 1300      TAD K7175        /-129 3 TIMES
2225 3443      DCA I WCLOC
2226 1074      TAD RBFLOC
2227 3444      DCA I CALOC
2230 4466      JMS I WAITI
2231 0001      1
2232 5253      JMP WDCERR      /NOT NORMAL INTERRUPT
2233 1443      TAD I WCLOC
2234 7640      SZA CLA
2235 5253      JMP WDCERR      /WC NOT ZERO
2236 1076      TAD DIRFLG
2237 7100      CLL
2240 7640      SZA CLA
2241 7120      STL
2242 1125      TAD K0003
2243 7430      SZL              /BACKWARDS IS -3
2244 7041      CMA IAC
2245 1040      TAD RECORD
2246 3040      DCA RECORD
2247 2277      ISZ TRECTR      /DONE ALL
2250 5223      JMP SWCMDL      /NO, DO NEXT 3 BLOCKS
2251 5652      JMP I .+1
2252 1503      WDINC

```

```

2253 3277 WDCERR,      DCA TRECTR      /FIND WHICH
2254 1443      TAD I WCLOC      /BLOCK WAS IN
2255 7041      CMA IAC          /ERROR
2256 1117      TAD K7577
2257 7510      SPA
2260 5253      JMP ,+3
2261 2277      ISZ TRECTR
2262 5256      JMP , -4
2263 7200      CLA
2264 1076      TAD DIRFLG
2265 7100      CLL
2266 7640      SZA CLA          /FORWARD
2267 7120      STL            /BACKWARD MAKE-1 OR -2

2270 1277      TAD TRECTR
2271 7430      SZL
2272 7041      CMA IAC
2273 1040      TAD RECORD
2274 3040      DCA RECORD      /BLOCK IN ERRJR
2275 5676      JMP I ,+1        /TYPEOUT WRITE ERROR
2276 1526      WSTERR

2277 0000      TRECTR,      0
2300 7175      <7175,      7175
2301 0150      <0150,      150

```

/READ ALL ROUTINE
/REV CKSUM DATA AND CKSUM ARE READ
/CKSUMS ARE GENERATED AND TESTED

*2400

2400	1076	RALLTS,	TAD DIRFLG	
2401	7450	SNA	/BACKWARDS	
2402	7001	IAC	/FORWARDS	
2403	7041	CMA IAC	/+1 FOR BACK-1 FOR FWD	
2404	1040	TAD RECORD		
2405	3040	DCA RECORD		
2406	1040	TAD RECORD		
2407	7510	SPA		
2410	5214	JMP ,+4		
2411	1147	TAD K5076		
2412	7710	SPA CLA		
2413	5217	JMP ,+4		
2414	7200	CLA		
2415	5616	JMP I ,+1	/CAVT DO 0 OR 2701	
2416	0235	MOFPR1		

/FIND THE BLOCK AND CHANGE TO READ ALL
/CLEAR READ ALL ERR FLG

2417	3377	DCA RAEFLG	/CLEAR READ ALL ERR. FL.	
2420	4455	JMS I SRCHIT		
2421	6764	IOT 764		
2422	1076	TAD DIRFLG		
2423	7450	SNA		
2424	7001	IAC		
2425	1040	TAD RECORD		
2426	3040	DCA RECORD		
2427	6773	IOT 773		
2430	5227	JMP , -1		
2431	7710	SPA CLA		
2432	5371	JMP RASERR		
2433	1041	TAD BLKFND		
2434	7041	CMA IAC		
2435	1040	TAD RECORD		
2436	7640	SZA CLA		
2437	5371	JMP RASERR		

/CHANGE TO READ ALL CONTINUOUS

2440	1374	TAD K0120	
2441	6764	IOT 764	
2442	1375	TAD K7571	
2443	3443	DCA I WCLOC	
2444	1075	TAD BF1LOC	
2445	3444	DCA I CALOC	
2446	4466	JMS I WAITI	
2447	0001	1	
2450	5770	JMP I RARERR	/READ ALL STATUS ERROR
2451	1443	TAD I WCLOC	
2452	7640	SZA CLA	
2453	5770	JMP I RARERR	
2454	1113	TAD K0200	
2455	6764	IOT 764	/STOP TAPE

/GENERATE CKSUMS AND TEST SUM=00

2456	1376	TAD RADLOC	
2457	3010	DCA 10	
2460	1117	TAD K7577	
2461	3011	DCA 11	
2462	3364	DCA CKSUMR	/CLR ACCUMULATED CKSUM
2463	1410	TAD I 10	
2464	0134	AND K0077	
2465	3363	DCA REVCHK	/SAVE REVERSE CKSUM
2466	1410	TAD I 10	
2467	4331	JMS XORSUM	/GENERATE DATA SUM
2470	2011	ISZ 11	
2471	5266	JMP , -3	
2472	1364	TAD CKSUMR	
2473	7040	CMA	
2474	0134	AND K0077	
2475	3361	DCA DATASM	/SAVE DATA SUM
2476	1410	TAD I 10	
2477	0127	AND K7700	
2500	3362	DCA FWDCHK	/SAVE FORWARD CHECKSUM
2501	1361	TAD DATASM	
2502	3364	DCA CKSUMR	
2503	1362	TAD FWDCHK	
2504	4331	JMS XORSUM	/DATA SUM + FORWARD CKECK
2505	1363	TAD REVCHK	
2506	4331	JMS XORSUM	/MAKE FINAL SJM
2507	1364	TAD CKSUMR	
2510	7640	SZA CLA	/SUM OK
2511	5760	JMP I CKSERR	/NO, ERROR
2512	1040	RADCHK, TAD RECORD	
2513	3451	DCA I POSITN	
2514	7040	CMA	
2515	1444	TAD I CALOC	
2516	3444	DCA I CALOC	
2517	4464	JMS I REGENP	
2520	4465	JMS I DATACO	
2521	6774	RANBFR	
2522	7202	BUFFRS+5	
2523	0201	0201	
2524	2377	ISZ RAEFLG	
2525	2072	ISZ COCNTR	
2526	5505	JMP I READ2A	
2527	5730	JMP I , +1	
2530	0217	MOFPRO-2	

/FORM 6 BIT XOR OF AC AND CKSUMR

/SAVE RESULTS IN CKSUMR

```

2531 5331 XORSUM, JMP ,
2532 7040 CMA
2533 3365 DCA XORSAY
2534 7040 CMA
2535 3366 DCA XORSAY+1 /TO COUNT PASSES
2536 1365 TAD XORSAY
2537 7040 CMA
2540 0364 AND CKSUMR
2541 3367 DCA XORSAY+2 /MAKE PARTIAL SUM
2542 1364 TAD CKSUMR
2543 7040 CMA
2544 0365 AND XORSAY /OTHER HALF SJM
2545 1367 TAD XORSAY+2 /COMBINE
2546 0134 AND K0077 /CLR TO LWR 6
2547 3364 DCA CKSUMR
2550 2366 ISZ XORSAY+1 /DONE UPPER 6?
2551 5731 JMP I XORSUM /DONE ALL EXIT
2552 1365 TAD XORSAY
2553 7012 RTR /MOVE OVER 6
2554 7012 RTR
2555 7012 RTR
2556 3365 DCA XORSAY
2557 5336 JMP XORSUM+5 /XOR UPPER

```

```

2560 2600 CKSERR, ERRCKS
2561 0000 DATASM, 0
2562 0000 FWDCHK, 0
2563 0000 REVCHK, 0
2564 0000 CKSUMR, 0
2565 0000 XORSAY, 0
2566 0000 0
2567 0000 0
2570 2707 RARERR, ERRRAR
2571 3151 RASERR, DCA PASFLG
2572 5773 JMP I .+1
2573 4000 SRHERR
2574 0120 <0120, 0120
2575 7571 <7571, 7571
2576 7200 RADLOC, BUFRS+3
2577 0000 RAEFLG, 0

```

/CHECK SUM ERROR TYPEOUT

*2600

2600	4701	ERRCKS,	JMS I CSECON	/TYRALL
2601	1071	TAD RECRDK		
2602	4506	JMS I TYPCON		
2603	4507	JMS I TYPTX		
2604	0042	42		
2605	5457	5457		
2606	4353	4353		
2607	7777	7777		
2610	4353	4353		
2611	6365	6365		
2612	5500	5500		
2613	4562	4562		
2614	6257	6257		
2615	6277	6277		
2616	1302	TAD CSECON+1	/REVCHK	
2617	4506	JMS I TYPCON		
2620	4507	JMS I TYPTX		
2621	0062	62		
2622	4566	4566		
2623	7700	7700		
2624	4257	JMS CKSTYP	/TYPE CHK SUM	
2625	1303	TAD CSECON+2	/DATASM	
2626	4506	JMS I TYPCON		
2627	4704	JMS I CSECON+3	/TYDATA	
2630	4257	JMS CKSTYP	/CHECKSUM	
2631	4270	JMS CALCTY		
2632	1305	TAD CSECON+4	/FWD CHECK	
2633	4506	JMS I TYPCON		
2634	4257	JMS CKSTYP	/CHK SUM	
2635	1706	TAD I CSECON+5		
2636	7040	CMA		
2637	0134	AND K0077		
2640	3706	DCA I CSECON+5		
2641	1306	TAD CSECON+5	/CKSUMR	
2642	4506	JMS I TYPCON		
2643	4270	JMS CALCTY		
2644	4507	JMS I TYPTX		
2645	0054	54	/L	
2646	1660	1660	/,P	
2647	1642	1642	/,B	
2650	1677	1677	/,	
2651	7040	CMA		
2652	3655	DCA I .+3		
2653	5654	JMP I .+1		
2654	2512	RADCHK		
2655	2577	RAEFLG		
2656	0525	REWDRV		

2657	5257	CKSTYP,	JMP ,
2660	4507	JMS I	TYPTX
2661	0043	43	
2662	5045	5045	
2663	4353	4353	
2664	0063	63	
2665	6555	6555	
2666	7700	7700	
2667	5657	JMP I	CKSTYP
2670	5270	CALCTY,	JMP ,
2671	4507	JMS I	TYPTX
2672	0043	43	
2673	4154	4154	
2674	4365	4365	
2675	5441	5441	
2676	6445	6445	
2677	4477	4477	
2700	5670	JMP I	CALCTY
2701	4432	CSECON,	TYRALL
2702	2563	REVCHK	
2703	2561	DATASM	
2704	4474	TYDATA	
2705	2562	FWDCHK	
2706	2564	CKSUMR	

/READ ALL STATUS ERROR TYPEOUT

2707	4467	ERRRAR, JMS I ERRSTP
2710	4701	JMS I CSECON
2711	1071	TAD RECDK
2712	4506	JMS I TYPCON
2713	4507	JMS I TYPTX
2714	0042	42
2715	5457	5457
2716	4353	4353
2717	7700	7700
2720	6772	10T 772
2721	3345	DCA ,+24
2722	1344	TAD ,+22
2723	4506	JMS I TYPCON
2724	4507	JMS I TYPTX
2725	0063	63
2726	6441	6441
2727	6400	6400
2730	4277	4277
2731	1043	TAD WCLOC
2732	4506	JMS I TYPCON
2733	4507	JMS I TYPTX
2734	0067	67
2735	1643	1643
2736	1677	1677
2737	7604	LAS
2740	0154	AND K6000
2741	7650	SNA CLA
2742	7402	HLT
2743	5656	JMP I CKSTYP=1
2744	2745	,+1
2745	0000	0

/TC01 RANDOM EXERCISER
/TYPEOUTS AND SOME CONTROL ROUTINES

*3000

/WAIT FOR INTERRUPT ROUTINE
/SYNCHRONIZE PROCESSOR TESTS

3000	5200	*ATINT,	JMP ,	
3001	1323	TAD WTKON1		/GET 5 SECOND WAIT K
3002	3325	DCA WTCNTR		
3003	6761	IOT 761		
3004	0123	AND K0070		
3005	7640	SZA CLA	/MOVE FUNCTION IN STAT	
3006	5211	JMP ,+3	/NO 5 SECONDS ENOUGH	
3007	1324	TAD WTKON2	/CHANGE WAIT COUNT TO	
3010	3325	DCA WTCNTR	/45 SECOND FOR MOVE	
3011	2162	ISZ FRSWAT	/FIRST PROGRAM WAIT	
3012	5217	JMP RESETD	/NO RESTORE AC AND LINK	
3013	4331	JMS SETIDX		
3014	6001	ION		
3015	5616	JMP I ,+1	/START IS TEST	
3016	3200	ISZTST		
3017	3162	RESETD,	DCA FRSWAT	/INDICATE NOT FIRST WAIT
3020	2342	ISZ PICNTR		
3021	7410	SKP		
3022	4331	JMS SETIDX		
3023	1000	TAD 0		
3024	3743	DCA I P1DEX1		
3025	2343	ISZ P1DEX1		
3026	1326	TAD LKSAVE		
3027	7010	RAR	/RESTORE LINK	
3030	7200	CLA		
3031	1327	TAD ACSAVE	/RESTORE ACCUMULATOR	
3032	6001	ION		
3033	5400	JMP I 0	/RETURN FROM LAST INTERRUPT	

/AFTER INTERRUPT RETURNS TO HERE

3034	3327	I RECD,	DCA ACSAVE	/SAVE ACCUMULATOR
3035	7004	RAL		
3036	3326	DCA	LKSAVE	/AND LINK FOR NEXT WAIT
3037	6771	IOT	771	
3040	5255	JMP	NDTSKP	/NO DECTAPE FLAG SET
3041	6772	IOT	772	
3042	3330	DCA	SBRECD	/SAVE STATUS B
3043	1600	TAD	I WATINT	
3044	7040	CMA		/GET STAT B COMPARE CON
3045	0330	AND	SBRECD	
3046	7650	SNA	CLA	/ANY ILLEGAL FLAGS
3047	2200	ISZ	WATINT	/NO IS SKIP EXIT TWICE
3050	2200	ISZ	WATINT	/ONLY 1 ISZ IF NOT I EXPECTED
3051	1200	TAD	WATINT	
3052	3744	DCA	I PDEX2	
3053	2344	ISZ	PDEX2	
3054	5600	JMP	I WATINT	

3055	4467	NDTSKP,	JMS I ERRSTP	/STOP TAPE
3056	4507	JMS	I TYPTX	
3057	7777		7777	
3060	6016		6016	
3061	5116		5116	
3062	0077		0077	
3063	5304	JMP	PTSTNS	

3064	3065	PTSTND,	.,+1	
3065	2325	ISZ	WTCNTR	/WAITED LONG ENOUGH
3066	5664	JMP	I PTSTND	/NO DO NEXT TEST
3067	6002	IOF		
3070	6771	IOT	771	/ANY FLAGS SET
3071	7040	CMA		/NO
3072	3330	DCA	SBRECD	
3073	4467	JMS	I ERRSTP	/STOP TAPE
3074	4507	JMS	I TYPTX	
3075	7777		7777	/NO PI
3076	5657		5657	
3077	0060		60	
3100	1651		1651	
3101	1677		1677	
3102	2330	ISZ	SBRECD	
3103	5307	JMP	.,+4	/DECTAPE DID SKIP

```

3104 4507 PTSTNS,      JMS I TYPTX
3105 0056          56
3106 5777          5777
3107 4507          JMS I TYPTX
3110 0044          44
3111 4543          4543
3112 6441          6441
3113 6045          6045
3114 0003          03
3115 5351          5351
3116 6077          6077
3117 2200          ISZ WATINT
3120 7040          CMA
3121 3162          DCA FRSWAT
3122 5600          JMP I WATINT

```

```

3123 7655 WTKON1,      7655      /WAIT COUNT 5 SECONDS
3124 6422 WTKON2,      6422      /WAIT COUNT 45 SECONDS
3125 0000 WTCNTR,      0
3126 0000 -KSAVE,      0
3127 0000 ACSAVE,      0
3130 0000 SBRECD,      0

```

```

3131 5331 SETIDX,      JMP .
3132 1341      TAD K7772Y
3133 3342      DCA PICNTR
3134 1345      TAD PITBL
3135 3343      DCA PIDEX1
3136 1354      TAD PIRTL
3137 3344      DCA PIDEX2
3140 5731      JMP I SETIDX

```

```

3141 7772 <7772Y,      7772
3142 0000 PICNTR,      0
3143 0000 PIDEX1,      0
3144 0000 PIDEX2,      0
3145 3146 PITBL,      .+1

```

```

      *PITBL+7
3154 3155 PIRTL,      .+1

```

Page 17
/PROCESS OR TEST FOR DECTREX1
/TESTS ARE RUN WHILE WAITING FOR INT

*3200

/ISZ TEST ABOUT 61 MILLISECONDS

3200	7040	ISZTST, CMA
3201	3345	DCA TEMP5
3202	3342	DCA TEMP2
3203	3341	DCA TEMP1
3204	2341	ISZ TEMP1
3205	2342	ISZ TEMP2
3206	5204	JMP , -2
3207	1342	TAD TEMP2
3210	7440	SZA
3211	7402	HLT
3212	7240	CLA CMA
3213	1341	TAD TEMP1
3214	7440	SZA
3215	7402	HLT
3216	2345	ISZ TEMP5
3217	7410	SKP
3220	5203	JMP ISZTST+3
3221	4750	JMS I NDPTST

/ROTATE 1 TEST ABOUT 67 MILLISECONDS

3222	1342	ROT1TS, TAD TEMP2
3223	7130	STL RAR
3224	7004	RAL
3225	7420	SNL
3226	7402	HLT
3227	7041	CMA IAC
3230	1342	TAD TEMP2
3231	7440	SZA
3232	7402	HLT
3233	2342	ISZ TEMP2
3234	5222	JMP ROT1TS
3235	4750	JMS I NDPTST

/ROTATE 2 TEST ALSO ABOUT 67 MILLISECONDS

3236	1342	ROT2TS,	TAD TEMP2
3237	7106	CLL RIL	
3240	7012	RTR	
3241	7430	SZL	
3242	7402	HLT	
3243	7041	CMA IAC	
3244	1342	TAD TEMP2	
3245	7440	SZA	
3246	7402	HLT	
3247	2342	ISZ TEMP2	
3250	5236	JMP ROT2TS	
3251	4750	JMS I NDPTST	

/TAD TEST ADD EVERY COM TO RAN VO
/ABOUT 86 MILLISECONDS

3252	3343	TADTST,	DCA TEMP3
3253	1346	TAD PRAN1	
3254	7104	CLL RAL	
3255	7430	SZL	
3256	7001	IAC	
3257	3346	DCA PRAN1	
3260	1347	TAD PRAN2	
3261	1346	TAD PRAN1	
3262	3347	DCA PRAN2	
3263	1347	TAD PRAN2	
3264	3344	DCA TEMP4	
3265	1347	TAD PRAN2	
3266	1343	TAD TEMP3	
3267	7041	CMA IAC	
3270	1344	TAD TEMP4	
3271	7440	SZA	
3272	7402	HLT	
3273	2344	ISZ TEMP4	
3274	7000	NOP	
3275	2343	ISZ TEMP3	
3276	5265	JMP .-11	
3277	4750	JMS I NDPTST	

/JMS TEST MAKE 13 PASSES OF 128 CONSECUTIVE JMS ,
/AND COMPARE RESULTS FOR ABOUT 63 MILLISECONDS

3300	1354	JMSTST,	TAD K7763X	
3301	3341	DCA	TEMP1	
3302	1353	TAD	K7600X	
3303	3342	DCA	TEMP2	
3304	1351	TAD	JMSLOC	
3305	3343	DCA	TEMP3	
3306	1352	TAD	JMSKON	
3307	3344	DCA	TEMP4	
3310	1344	TAD	TEMP4	/STORE 128 JMS .
3311	3743	DCA I	TEMP3	/STARTING AT ADDRESS
3312	2344	ISZ	TEMP4	/6000
3313	2343	ISZ	TEMP3	
3314	2342	ISZ	TEMP2	
3315	5310	JMP	,-5	
3316	1355	TAD	RETJMP	/STORE JMP I RETJMP
3317	3743	DCA I	TEMP3	/TO RETURN FROM JMS
3320	5751	JMP I	JMSLOC	/EXECUTE 128 JMS
3321	1353	JMRETU,	TAD K7600X	/RETURN FROM EXECUTE
3322	3342	DCA	TEMP2	
3323	1351	TAD	JMSLOC	
3324	3343	DCA	TEMP3	/COMPARE ADDRESSES
3325	1343	TAD	TEMP3	/FOR .+1
3326	7040	CMA		
3327	1743	TAD I	TEMP3	
3330	7440	SZA		
3331	7402	HLT		
3332	2343	ISZ	TEMP3	/INC COMP AND FETCH
3333	2342	ISZ	TEMP2	/DONE 128 YET
3334	5325	JMP	JMRETU+4	
3335	2341	ISZ	TEMP1	
3336	5302	JMP	JMSTST+2	
3337	4750	JMS I	NDPTST	
3340	5200	JMP	ISZTST	
3341	0000	TEMP1,	0	
3342	0000	TEMP2,	0	
3343	0000	TEMP3,	0	
3344	0000	TEMP4,	0	
3345	0000	TEMP5,	0	
3346	4263	PRAN1,	4263	
3347	2634	PRAN2,	2634	
3350	3064	NDPTST,	PTSTND	
3351	6000	JMSLOC,	6000	
3352	4200	JMSKON,	4200	/JMS + 200
3353	7600	K7600X,	7600	/FOR COUNTING JMS
3354	7763	K7763X,	7763	
3355	5561	RETJMP,	JMP I RETJMP	/(RETJMP) ARE JMRETU

PAUSE

/TC01 DECTEX 1 - TAPE 3
 /REWIND ALL DRIVES SELECTED
 /TO END ZONE AT START OF TAPE

*3400

```

3400 5200 REPOS1,      JMP ,
3401 4217      JMS RSFDRV      /RESET POINTERS TO FIRST DRIVE
3402 1140      TAD K0604      /MOVE BACKWARDS
3403 1046      TAD UNFUNC      /*POSITIONED UNIT NO
3404 6766      IOT 766      /SET STATUS A
3405 4466      JMS I WAITI
3406 5000      5000      /INDICATE EXPECT END
3407 5340      JMP MOVER      /NO! STATUS EXPECTED

3410 7240      CLA CMA
3411 3451      DCA I POSITN      /INDICATE END ZONE
3412 7240      CLA CMA
3413 3452      DCA I DIRECT      /INDICATE BACKWARDS
3414 4237      JMS CHNGDR      /SET UP NEXT DRIVE
3415 5202      JMP REPOS1+2      /REWIND NEXT DRIVE
3416 5600      JMP I REPOS1      /GOT ALL DRIVES, EXIT

```

/RESET CURRENT DRIVE POINTERS TO
 /FIRST DRIVE SELECTED

```

3417 5217 RSFDRV,      JMP ,
3420 7200      CLA
3421 3045      DCA CDRIVE      /SET INITIALLY TO 0
3422 1121      TAD K4000
3423 3050      DCA COMBIT
3424 1047      TAD MSBITS

3425 0050      AND COMBIT
3426 7640      SZA CLA      /THIS DRIVE SELECTED
3427 5235      JMP RSFDR1      /YES, SET POINTER
3430 1050      TAD COMBIT
3431 7110      CLL RAR
3432 3050      DCA COMBIT      /MOVE COMPARE BIT
3433 2045      ISZ CDRIVE      /INCREMENT DRIVE NUM.
3434 5224      JMP RSFDRV+5

```

/HAVE FOUND FIRST DRIVE SELECTED

```

3435 4261 RSFDR1,      JMS GNPTRS      /GENERATE CONTROL POINTERS
3436 5617      JMP I RSFDRV      /EXIT

```

/SELECT NEXT DRIVE OR
/RESET TO FIRST DRIVE AND SKIP
CHNGDR, JMP .

3437	5237		
3440	7200	CLA	
3441	1050	TAD COMBIT	/GET DRIVE COMPARE BIT
3442	7110	CLL RAR	/MOVE IT TO NEXT
3443	0300	AND NBIT8	
3444	7440	SZA	/LAST DRIVE NJM 7
3445	5251	JMP ,+4	/NO
3446	4217	JMS RSFDRV	/RESET TO FIRST
3447	2237	ISZ CHNGDR	/INCR. EXIT, SKIP
3450	5637	JMP I CHNGDR	/EXIT
3451	3050	DCA COMBIT	
3452	2045	ISZ CDRIVE	
3453	1050	TAD COMBIT	
3454	0047	AND MSBITS	
3455	7650	SNA CLA	/THIS DRIVE SELECTED
3456	5240	JMP CHNGDR+1	/NO
3457	4261	JMS GNPTRS	/GENERATE DRIVE POINTERS
3460	5637	JMP I CHNGDR	

/GENERATE LAST RECMRD,
/DIRECTION AND UNIT NUMBER POINTERS
/FOR DECTAPE FUNCTIONS

3461	5261	GNPTRS, JMP .	
3462	1045	TAD CDRIVE	/DRIVE NUMBER
3463	7112	CLL RTR	
3464	7012	RTR	/POSITION TO BITS 0,1,2
3465	3046	DCA UNFUNC	
3466	1045	TAD CDRIVE	/DRIVE NUMBER
3467	1301	TAD PNTRS	/+ POS. PNTR ADDRS,
3470	3051	DCA POSITN	/FOR INDIRECTS
3471	1045	TAD CDRIVE	
3472	1312	TAD PNTRS+11	/+ DIRECTION PNTR
3473	3052	DCA DIRECT	/FOR INDIRECTS
3474	1045	TAD CDRIVE	
3475	1323	TAD PNTRS+22	
3476	3053	DCA LSTBLK	
3477	5661	JMP I GNPTRS	
3500	7767	NBIT8, 7767	

3501	3502	PNTRS,	.+1	/TO GET LAST RECORD NUMBER
3502	0000	0		/FOR DRIVE 8
3503	0000	0		/1
3504	0000	0		/2
3505	0000	0		/3
3506	0000	0		/4
3507	0000	0		/5
3510	0000	0		/6
3511	0000	0		/7
3512	3513	.+1		/TO GET LAST DIRECTION
3513	0000	0		/DIRECTION - UNIT 8
3514	0000	0		/1
3515	0000	0		/2
3516	0000	0		/3
3517	0000	0		/4
3520	0000	0		/5
3521	0000	0		/6
3522	0000	0		/7
3523	3524	.+1		
3524	0000	0		/TO GET LAST BLOCK WRITTEN
3525	0000	0		/DRIVE 8
3526	0000	0		/1
3527	0000	0		/2
3530	0000	0		/3
3531	0000	0		/4
3532	0000	0		/5
3533	0000	0		/6
3534	0000	0		/7
3535	3536	.+1		
3536	0000	0		
3537	4400	TYMOVE		
3540	4342	MOVER, JMS .+2		
3541	5202	JMP REPOS+2		
3542	5342	JMP ,		
3543	4467	JMS I ERRSTP		
3544	4737	JMS I MOVER-1	/TYMOVE	
3545	6772	IOT 772		
3546	3336	DCA MOVER-2		
3547	1335	TAD MOVER-3		
3550	4506	JMS I TYPCON		
3551	4507	JMS I TYPTX		
3552	0063	63		
3553	6441	6441		
3554	6400	6400		
3555	4277	4277		
3556	7624	LAS		
3557	0154	AND K6000		
3560	7650	SNA CLA		
3561	7402	HLT		
3562	5742	JMP I MOVER+2		

/SEARCH ROUTINE FIND
 /BLOCK IN (RECORD) IN DIRECTION
 /INDICATED BY DIRFLG==0 FWD
 /=7777 BKWD

*3600

3600	5200	SEARCH,	JMP .	
3601	7300	CLL CLA		
3602	1076	TAD DIRFLG		
3603	7640	SZA CLA	/FORWARD	
3604	7120	STL	/NO, BACKWARD	
3605	1125	TAD K0003	/BACKWARD TA	
3606	7420	SNL	/IS BLOCK+3	
3607	7041	CMA IAC	/FWD IS BLOCK-3	
3610	1040	TAD RECORD		
3611	7041	CMA IAC		
3612	3156	DCA TAPONT	/INDICATE TA POINT	
3613	1076	TAD DIRFLG		
3614	7650	SNA CLA	/FORWARD IS	
3615	1137	TAD K0400	/START BKWD	
3616	1126	TAD K0214		
3617	1046	TAD UNFUNC		
3620	6766	IOT 766		
3621	1042	TAD IDCON		
3622	3444	DCA I CALOC		
3623	7040	CMA		
3624	3157	DCA BLKFLG		
3625	4466	JMS I WAITI		
3626	0001	1		
3627	5326	JMP SREZTS	/SEE IF EZ FOUND	
3630	4263	JMS SRCONT		
3631	5235	JMP SRTARN	/BLOCK=BLOCKFND	
3632	5235	JMP SRTARN	/GONE PAST BLK	
3633	6764	IOT 764	/HAVEN'T REACHED BLOCK	
3634	5225	JMP , -7	/FOR TURN AROUND YET	

```

3635 6761 SRTARN, IOT 761
3636 7040 CMA
3637 0113 AND K0200 /IN CASE MOTION=0
3640 1137 TAD K0400
3641 6764 IOT 764 /DO TURN AROUND
3642 1040 TAD RECORD
3643 7041 CMA IAC
3644 3156 DCA TAPONT
3645 7040 CMA
3646 3157 DCA BLKFLG
3647 4466 JMS I WAITI
3650 0021 1
3651 5555 JMP I SRCHER
3652 4263 JMS SRCONT /CHK RELATION OF BLOCK
3653 5620 JMP I SEARCH /FOUND BLOCK, EXIT
3654 5555 JMP I SRCHER /GONE PAST, ERROR
3655 7040 CMA
3656 1137 TAD BLKFLG
3657 7650 SNA CLA /READ 2 BLOCK NUMS
3660 5355 JMP SRCMOD /YES CHNG TO C MODE
3661 6764 IOT 764 /HAVEN'T REACHED YET
3662 5247 JMP , -13 /WAIT FOR NEXT

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```

/FIND RELATION OF BLOCK FOUND
/TO BLOCK SOUGHT AND TEST
/BLOCK NUMBERS TO BE CONSECUTIVE
SRCONT, JMP .

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```

3663 5263
3664 2157 ISZ BLKFLG /FIRST BLOCK NUMBER
3665 7410 SKP /NOT FIRST
3666 5302 JMP SRFBK /FIRST, JUST SAVE IT
3667 6761 IOT 761
3670 0137 AND K0400
3671 7640 SZA CLA /BKWRDS IS
3672 7040 CMA /-1

3673 7450 SNA /FWD IS
3674 7001 IAC /+1
3675 1160 TAD PREBLK /LAST BLOCK +JR-1
3676 7041 CMA IAC
3677 1041 TAD BLKFND
3700 7640 SZA CLA /BLOCKS CONSECUTIVE
3701 5555 JMP I SRCHER /NO, ERROR

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3702	1041	SRFBLK,	TAD BLKFND	
3703	3160	DCA	PREBLK	
3704	1041	TAD	BLKFND	
3705	1156	TAD	TAPONT	
3706	7450	SNA		/BLOCKS=
3707	5663	JMP I	SRCONT	/YES, FOUND IT
3710	2263	ISZ	SRCONT	
3711	7120	CLL		
3712	7710	SPA	CLA	/BLKFND GREATER
3713	7120	STL		/NO, LESS
3714	6761	IOT	761	
3715	0137	AND	K0400	
3716	7430	SZL		
3717	5323	JMP	,+4	
3720	7640	SZA	CLA	/IF FWD HAVE GONE PAST
3721	2263	ISZ	SRCONT	/BKWD
3722	5663	JMP I	SRCONT	
3723	7650	SNA	CLA	
3724	2263	ISZ	SRCONT	
3725	5663	JMP I	SRCONT	
3726	6772	SREZTS,	IOT 772	
3727	0131	AND	EZBIT	
3730	7650	SNA	CLA	/END ZONE INTERRUPT
3731	5555	JMP I	SRCHER	/NO, SOME OTHER ERROR
3732	1156	TAD	TAPONT	
3733	7500	SMA		/BLOCK 0 OR 1
3734	5340	JMP	,+4	/YES HIT EZ AGAIN
3735	1153	TAD	K2701	
3736	7700	SMA	CLA	/BLOCK 2701 OR 2700
3737	5352	JMP	,+13	/NO, TURN AROUND
3740	7604	LAS		
3741	7010	RAR		
3742	7620	SNL	CLA	/NEW FORMAT TAPE
3743	5235	JMP	SRTARN	/YES TURN AROUND
3744	1113	TAD	K0200	/SET MOTION AGAIN
3745	6764	IOT	764	
3746	4466	JMS I	WAITI	/WAIT FOR EZ AGAIN
3747	5000		5000	
3750	5555	JMP I	SRCHER	
3751	5235	JMP	SRTARN	
3752	2157	ISZ	BLKFLG	/NOT FIRST INTERRUPT
3753	5555	JMP I	SRCHER	/EZ IS ERROR
3754	5235	JMP	SRTARN	/TURN AROUND

3755	1041	SRCMOD,	TAD BLKFND	/FIND DIFFERENCE
3756	7041	CMA IAC	/IV NUMBER OF BLOCKS	
3757	1040	TAD RECORD		
3760	7500	SMA		
3761	7041	CMA IAC	/MAKE -	
3762	3443	DCA I WCLOC	/FOR WORD COUNT	
3763	1135	TAD K0100		
3764	6764	IOT 764	/SET CONTIN MODE	
3765	4466	JMS I WAITI		
3766	0001	1		
3767	5555	JMP I SRCHER	/NOT NORMAL INTERRUPT	
3770	1041	TAD BLKFND		
3771	7041	CMA IAC		
3772	1040	TAD RECORD		
3773	7640	SZA CLA	/RIGHT BLOCK	
3774	5555	JMP I SRCHER	/NO. ERROR	
3775	1135	TAD K0100	/CLEAR CONTIN MODE	
3776	6764	IOT 764		
3777	5600	JMP I SEARCH		

/SEARCH ERROR TYPEOUT

*4000

4000	4467	SRHERR.	JMS I ERRSTP
4001	4545		JMS I SRCHTY
4002	1071		TAD RECRDK
4003	4506		JMS I TYPCON
4004	4507		JMS I TYPTX
4005	0042		42
4006	5457		5457
4007	4353		4353
4010	0067		67
4011	4156		4156
4012	6445		6445
4013	4400		4400
4014	7700		7700
4015	1076		TAD DIRFLG
4016	7650	SNA	CLA
4017	5222		JMP ,+3
4020	4714		JMS I BACKTY
4021	7410		SKP
4022	4715		JMS I FORDTY
4023	1042		TAD IDCON
4024	4506		JMS I TYPCON
4025	4507		JMS I TYPTX
4026	0042		42
4027	5457		5457
4030	4353		4353
4031	0046		46
4032	5765		5765
4033	5644		5644
4034	7700		7700
4035	7040		CMA
4036	1710		TAD I SEKONS
4037	7710		SPA CLA
4040	5252		JMP ,+12

/BLKF-G

4041	1311	TAD SEKONS+1	/PREBLK
4042	4506	JMS I TYPCON	
4043	4507	JMS I TYPTX	
4044	0054	54	
4045	4163	4163	
4046	6400	6400	
4047	4254	4254	
4050	5743	5743	
4051	5377	5377	
4052	2710	ISZ I SEKONS	
4053	7000	NOP	
4054	1310	TAD SEKONS	
4055	4506	JMS I TYPCON	
4056	4507	JMS I TYPTX	
4057	0042	42	
4060	5457	5457	
4061	4353	4353	
4062	6300	6300	
4063	6245	6245	
4064	4144	4144	
4065	7700	7700	
4066	6772	IOT 772	
4067	3313	DCA SEKONS+3	
4070	1312	TAD SEKONS+2	
4071	4506	JMS I TYPCON	
4072	4507	JMS I TYPTX	
4073	0063	63	
4074	6441	6441	
4075	6400	6400	
4076	4277	4277	
4077	7604	LAS	
4100	0154	AND K6000	
4101	7650	SNA CLA	
4102	7402	HLT	
4103	2151	ISZ PASFLG	/2ND ERROR
4104	5707	JMP I .+3	/REWIND DRIVE
4105	5706	JMP I .+1	
4106	3601	SEARCH+1	
4107	0525	REWDRV	
4110	0157	SEKONS, BLKFLG	
4111	0160	PREBLK	
4112	4113	.+1	
4113	0000	0	
4114	4547	BACKTY, TYBKW	
4115	4557	FORDTY, TYFWD	

/TYPE TEXT ROUTINE

*4200

4200	5200	TYTEXT, JMP ,	
4201	7200	CLA	
4202	1600	TAD I TYTEXT	/GET NEXT 2 CHARACTERS
4203	7040	CMA	/MAKE -
4204	3320	DCA TXSTOR	
4205	2200	ISZ TYTEXT	
4206	1320	TAD TXSTOR	
4207	7440	SZA	/CARRIAGE RETURN - LINE FEED
4210	5213	JMP ,+3	/NO
4211	4250	JMS CRLFLF	/CR LF
4212	5201	JMP TYTEXT+1	/GET NEXT
4213	0127	AND K7700	/CLEAR TO UPR CHAR
4214	7450	SNA	/END OF MESSAGE
4215	5600	JMP I TYTEXT	/YES
4216	7012	RTR	/MOVE
4217	7012	RTR	/OVER
4220	7012	RTR	/6 PLACES
4221	4230	JMS TYCHAR	/OUTPUT
4222	1320	TAD TXSTOR	
4223	0134	AND K0077	
4224	7450	SNA	/END OF MESSAGE
4225	5600	JMP I TYTEXT	/YES EXIT
4226	4230	JMS TYCHAR	/OUTPUT
4227	5201	JMP TYTEXT+1	/GET NEXT 2
4230	5230	TYCHAR, JMP ,	
4231	3247	DCA CRLFLF-1	
4232	7604	LAS	
4233	7710	SPA CLA	
4234	5630	JMP I TYCHAR	
4235	1247	TAD CRLFLF-1	
4236	7040	CMA	/MAKE + AGAIN
4237	0134	AND K0077	/CLEAR TO LOWER 6
4240	1136	TAD K0240	/MAKE ASCII
4241	6046	TLS	/OUTPUT
4242	6041	TSF	/WAIT FLAG
4243	5242	JMP , -1	
4244	7200	CLA	
4245	6042	TCF	/CLEAR FLAG
4246	5630	JMP I TYCHAR	/DO NEXT

4247	0000	0	
4250	5250	CRLFLF,	JMP ,
4251	7604	LAS	
4252	7710	SPA CLA	/DELETE TYPEOJTS
4253	5650	JMP I CRLFLF	
4254	1270	TAD K0215	/CARRIAGE RETURN
4255	6046	TLS	
4256	6041	TSF	
4257	5256	JMP , -1	
4260	7200	CLA	
4261	1271	TAD K0212	/LINE FEED
4262	6046	TLS	
4263	6041	TSF	
4264	5263	JMP , -1	
4265	6042	TCF	
4266	7200	CLA	
4267	5650	JMP I CRLFLF	
4270	0215	0215,	215
4271	0212	0212,	212

Page 00
/TYPE CONTENTS OF ADDRESS IN AC

```

4272 5272 TYCONT,    JMP ,
4273 3320      DCA TXSTOR      /SAVE ADDRESS
4274 1720      TAD I TXSTOR    /GET CONTENTS
4275 3320      DCA TXSTOR
4276 4250      JMS CRLFLF      /CARRIAGE RETURN - LINE FEED
4277 4304      JMS TYCOVR      /TYPE UPPER OCTAL
4300 4304      JMS TYCOVR
4301 4304      JMS TYCOVR
4302 4304      JMS TYCOVR
4303 5672      JMP I TYCONT

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4304 5304 TYCOVR,    JMP ,
4305 1320      TAD TXSTOR
4306 7006      RTL
4307 7004      RAL
4310 3320      DCA TXSTOR
4311 1320      TAD TXSTOR
4312 7004      RAL
4313 0120      AND K0007
4314 1133      TAD K0020
4315 7040      CMA              /MAKE - FOR
4316 4230      JMS TYCHAR      /OUTPUT
4317 5704      JMP I TYCOVR
4320 0000 TXSTOR,    0

```

```

/STOP TAPE ON ERROR, LEAVE FLAGS SET
4321 5321 ERSTP,    JMP ,
4322 6761      IOT 761
4323 0113      AND K0200
4324 1125      TAD K0003
4325 6764      IOT 764
4326 5721      JMP I ERSTP

```

*4400

/TYPE MOVE AND DIRECTION

4400	5000	TYMOVE, JMP
4401	4311	JMS TYDRV
4402	4507	JMS I TYPTX
4403	7777	7777
4404	5557	5557
4405	6645	6645
4406	0077	0077
4407	4340	JMS TYDIR
4410	5600	JMP I TYMOVE

/TYPE SEARCH DIRECTION AND MODE

4411	5211	TYSRCH, JMP
4412	4311	JMS TYDRV
4413	4507	JMS I TYPTX
4414	6345	6345
4415	4162	4162
4416	4350	4350
4417	0077	0077
4420	4340	JMS TYDIR
4421	4365	JMS TYMODE
4422	5611	JMP I TYSRCH

/TYPE READ DATA DIRECTION AND MODE

4423	5223	TYRDAT, JMP
4424	4311	JMS TYDRV
4425	4257	JMS TYREAD
4426	4274	JMS TYDATA
4427	4340	JMS TYDIR
4430	4365	JMS TYMODE
4431	5623	JMP I TYRDAT

/TYPE READ ALL DIRECTION AND MODE

4432	5232	TYRALL, JMP
4433	4311	JMS TYDRV
4434	4257	JMS TYREAD
4435	4303	JMS TYALL
4436	4340	JMS TYDIR
4437	4365	JMS TYMODE
4440	5632	JMP I TYRALL

/TYPE WRITE DATA DIRECTION AND MODE

```
4441 5241 TYWDAT, JMP .
4442 4311 JMS TYDRV
4443 4265 JMS TYWRIT
4444 4274 JMS TYDATA
4445 4340 JMS TYDIR
4446 4365 JMS TYMODE
4447 5641 JMP I TYWDAT
```

/TYPE WRITE ALL DIRECTION AND MODE

```
4450 5250 TYWALL, JMP .
4451 4311 JMS TYDRV
4452 4265 JMS TYWRIT
4453 4303 JMS TYALL
4454 4340 JMS TYDIR
4455 4365 JMS TYMODE
4456 5650 JMP I TYWALL
```

/TYPE READ

```
4457 5257 TYREAD, JMP .
4460 4507 JMS I TYPTX
4461 6245 6245
4462 4144 4144
4463 0077 0077
4464 5657 JMP I TYREAD
```

/TYPE WRITE

```
4465 5265 TYWRIT, JMP .
4466 4507 JMS I TYPTX
4467 6762 6762
4470 5164 5164
4471 4500 4500
4472 7700 7700
4473 5665 JMP I TYWRIT
```

/TYPE DATA

```
4474 5274 TYDATA, JMP .
4475 4507 JMS I TYPTX
4476 0000 0
4477 4441 4441
4500 6441 6441
4501 0077 0077
4502 5674 JMP I TYDATA
```

```

/TYPE ALL
4503 5303 TYALL, JMP ,
4504 4507 JMS I TYPTX
4505 4154 4154
4506 5400 5400
4507 7700 7700
4510 5703 JMP I TYALL

```

```

/TYPE DRIVE AND NUMBER
4511 5311 TYDRV, JMP ,
4512 7604 LAS
4513 7710 SPA CLA
4514 5711 JMP I TYDRV
4515 4507 JMS I TYPTX
4516 7777 7777
4517 7777 7777
4520 4462 4462
4521 5166 5166
4522 4500 4500
4523 0077 0077
4524 1045 TAD CORIVE
4525 7450 SNA
4526 1132 TAD K0010
4527 1142 TAD K260
4530 6046 TLS
4531 6041 TSF
4532 5331 JMP , -1
4533 6042 TCF
4534 4507 JMS I TYPTX
4535 7777 7777
4536 7700 7700
4537 5711 JMP I TYDRV

```

```

/TYPE FORWARDS OR BACKWARD
4540 5340 TYDIR, JMP ,
4541 6761 IOT 761
4542 0137 AND K0400
4543 7650 SNA CLA
4544 5355 JMP TYFWD-2
4545 4347 JMS TYBKW
4546 5740 JMP I TYDIR
4547 5347 TYBKW, JMP ,
4550 4507 JMS I TYPTX
4551 4253 4253
4552 6744 6744
4553 0077 0077
4554 5747 JMP I TYBKW

```

```

4555 4357 JMS TYFWD
4556 5740 JMP I TYDIR
4557 5357 TYFWD, JMP ,
4560 4507 JMS I TYPTX
4561 4667 4667
4562 4400 4400
4563 7700 7700

```

/TYPE CONTINUOUS IF NOT NORMAL MODE

4565	5365	TYMODE, JMP
4566	6761	IOT 761
4567	0135	AND K0100
4570	7650	SNA CLA
4571	5765	JMP I TYMODE
4572	4507	JMS I TYPTX
4573	4300	4300
4574	5557	5557
4575	4445	4445
4576	0077	0077
4577	5765	JMP I TYMODE

ACSAVE	3127
BACKTY	4114
BF1LOC	0075
BF2LOC	0103
BLKFLG	0157
BLKFND	0041
BUFFRS	7175
CALCTY	2670
CALOC	0044
CDRIVE	0045
CHNGDR	3437
CKSERR	2560
CKSTYP	2657
CKSUMR	2564
COCNTR	0072
CODATA	1225
CODEX	1255
COERRO	1261
COERR1	1313
COLOOP	1243
COMBIT	0050
COMSTR	0755
CORT1R	1442
CRLFLF	4250
CSECON	2701
DATA00	0065
DATAMV	0070
DATASM	2561
DIRECT	0052
DIRFLG	0076
DIRTBL	1044
DRIVTY	0130
ERRCKS	2600
ERRRAR	2707
ERRSTP	0067
ERSTP	4321
EXIST	1025
EZBIT	0131
FORDTY	4115
FRSWAT	0102
FWDCHK	2562
GENDEX	0675
GENPAT	0061
GENRAN	0735

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ACSAVE 3127
 BACKTY 4114
 BF1LOC 0075
 BF2LOC 0103
 BLKFLG 0157
 BLKFND 0041
 BUFFRS 7175
 CALCTY 2670
 CALOC 0044
 CDRIVE 0045
 CHNGDR 3437
 CKSERR 2560
 CKSTYP 2657
 CKSUMR 2564
 COCNTR 0072
 CODATA 1225
 CODEX 1255
 COERRO 1261
 COERR1 1313
 COLOOP 1243
 COMBIT 0050
 COMSTR 0755
 CORT1R 1442
 CRLFLF 4250
 CSECON 2701
 DATACO 0065
 DATAMV 0070
 DATASM 2561
 DIRECT 0052
 DIRFLG 0076
 DIRTBL 1044
 DRIVTY 0130
 ERRCKS 2600
 ERRRAR 2707
 ERRSTP 0067
 ERSTP 4321
 EXIST 1025
 EZBIT 0131
 FORDTY 4115
 FRSWAL 0162
 FWDCHK 2562
 GENDEX 0675
 GENPAT 0061
 GENRAN 0735
 GETRED 0570
 GNPATR 0626
 GNPTRS 3461
 IDCON 0042
 IOT 6000
 IRECD 3034
 ISZTST 3200
 JMRETU 3321
 JMSKON 3352
 JMSLOC 3351
 JMSTST 3300
 K0002 0152
 K0003 0125
 K0007 0120

K0010 0132
 K0020 0133
 K0030 0116
 K0037 0111
 K0040 0114
 K0050 0150
 K0070 0123
 K0077 0134
 K0100 0135
 K0120 2574
 K0130 2113
 K0150 2301
 K0200 0113
 K0212 4271
 K0214 0126
 K0215 4270
 K0240 0136
 K0400 0137
 K0604 0140
 K0614 0115
 K0700 0122
 K1000 0131
 K16 0141
 K260 0142
 K2701 0153
 K3777 0143
 K4000 0121
 K5075 0146
 K5076 0147
 K6000 0154
 K7000 0124
 K7175 2300
 K7571 2575
 K7577 0117
 K7600 0543
 K7600X 3353
 K7700 0127
 K7760 0110
 K7763X 3354
 K7770 0112
 K7772Y 3141
 K7774 1354
 LKSAVE 3126
 LSTBLK 0053
 LSTDRV 0054
 LSTTBL 1045
 MCOMOB 0703
 MIN4 0753
 MIN60 0754
 MOFPRO 0221
 MOFPR1 0235
 MOVDEX 1222
 MOVER 3540
 MSBITS 0047
 MVDATA 1200
 NBIT8 3500
 NDOFRD 1725
 NDPTST 3350
 NDTSKP 3055
 NEWDRV 0057

NUMBLK 0077
 PASFLG 0151
 PATGEN 0600
 PICNTR 3142
 PIDEX1 3143
 PIDEX2 3144
 PIRTBL 3154
 PITBL 3145
 PNTRS 3501
 POSITN 0051
 POSTBL 1043
 PRAN1 3346
 PRAN2 3347
 PREBLK 0160
 PREGEN 0547
 PTSTND 3064
 PTSTNS 3104
 RADCHK 2512
 RADLOC 2576
 RAFLG 2577
 RALLTS 2400
 RANBFR 6774
 RANDEX 0200
 RANGEN 0062
 RANNO 0751
 RANSEL 1000
 RANVAR 0752
 RARERR 2570
 RASERR 2571
 RBFLOC 0074
 RBFWD2 0102
 RBUFND 0100
 RBUFST 0073
 RDBUF1 1632
 RDCERR 2075
 RDCMOD 2000
 RDDATA 0400
 RDSEQ 1600
 RDTAB 0304
 READTY 0144
 READ1 0063
 READ2 0104
 READ2A 0105
 RECORD 0040
 RECRDK 0071
 REGENP 0064
 REPOSI 3400
 REREAD 0420
 REREDA 0470
 RESETD 3017
 RETJMP 3355
 RETUJM 0161
 REVCHK 2563
 REWCK 1517
 REWDRV 0525
 REWIND 0056
 ROT1TS 3222
 ROT2TS 3236
 RPATO 0700
 RSFDRV 3417

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RSFDR1 3435
 RSQFLG 0350
 SAMDRV 0325
 SAVNUM 1570
 SBRECD 3130
 SEARCH 3600
 SEKONS 4110
 SELGEN 1046
 SELRAN 0060
 SELVAR 1061
 SETIDX 3131
 SRCHER 0155
 SRCHIT 0055
 SRCHTY 0145
 SRCMOD 3755
 SRCONT 3663
 SREZTS 3726
 SRFBLK 3702
 SRHERR 4000
 SRTARN 3635
 SWCMDL 2223
 SWCMOD 2212
 TADTST 3252
 TAPONT 0156
 TEMP1 3341
 TEMP2 3342
 TEMP3 3343
 TEMP4 3344
 TEMP5 3345
 TRECTR 2277
 TXSTOR 4320
 TYALL 4503
 TYBKW 4547
 TYCHAR 4230
 TYCONT 4272
 TYCOVR 4304
 TYDATA 4474
 TYDIR 4540
 TYDRV 4511
 TYFWD 4557
 TYMODE 4565
 TYMOVE 4400
 TYPCON 0106
 TYPTX 0107
 TYRALL 4432
 TYRDAT 4423
 TYREAD 4457
 TYSRCH 4411
 TYTEXT 4200
 TYWALL 4450
 TYWDAT 4441
 TYWRIT 4465
 UNFUNC 0046
 WAITI 0066
 WATINT 3000
 WCLOC 0043
 WDCERR 2253
 WDCMOD 2200
 WDINC 1503
 WRITES 0324

GNPATR 0626
 GNPTRS 3461
 IDCON 0042
 IOT 6000
 IRECD 3034
 ISZTST 3200
 JMRETU 3321
 JMSKON 3352
 JMSLOC 3351
 JMSTST 3300
 K0002 0152
 K0003 0125
 K0007 0120
 K0010 0132
 K0020 0133
 K0030 0116
 K0037 0111
 K0040 0114
 K0050 0150
 K0070 0123
 K0077 0134
 K0100 0135
 K0120 2574
 K0130 2113
 K0150 2301
 K0200 0113
 K0212 4271
 K0214 0126
 K0215 4270
 K0240 0136
 K0400 0137
 K0604 0140
 K0614 0115
 K0700 0122
 K1000 0131
 K16 0141
 K260 0142
 K2701 0153
 K3777 0143
 K4000 0121
 K5075 0146
 K5076 0147
 K6000 0154
 K7000 0124
 K7175 2300
 K7571 2575
 K7577 0117
 K7600 0543
 K7600X 3353
 K7700 0127
 K7760 0110
 K7763X 3354
 K7770 0112
 K7772Y 3141
 K7774 1354
 LKSAVE 3126
 LSTBLK 0053
 LSTDRV 0054
 LSTTBL 1045
 MCOMOB 0703

MIN4 0753
 MIN60 0754
 MOFPRO 0221
 MOFPR1 0235
 MOVDEX 1222
 MOVER 3540
 MSBITS 0047
 MVDATA 1200
 NBIT8 3500
 NDOFRD 1725
 NDPTST 3350
 NOTSKP 3055
 NEWDRV 0057
 NUMBLK 0077
 PASFLG 0151
 PATGEN 0600
 PICNTR 3142
 PIDEK1 3143
 PIDEK2 3144
 PIRTL 3154
 PITBL 3145
 PNTRS 3501
 POSITN 0051
 POSTBL 1043
 PRAN1 3346
 PRAN2 3347
 PREBLK 0160
 PREGEN 0547
 PTSTND 3064
 PTSTNS 3104
 RADCHK 2512
 RADLOC 2576
 RAEFLG 2577
 RALLTS 2420
 RANBFR 6774
 RANDEX 0200
 RANGEN 0062
 RANN0 0751
 RANSEL 1000
 RANVAR 0752
 RARERR 2570
 RASERR 2571
 RBFLOC 0074
 RBFWD2 0102
 RBUFND 0100
 RBUFST 0073
 RDBUF1 1632
 RDCERR 2075
 RDCMOD 2000
 RDDATA 0400
 RDSEQ 1600
 RDTAB 0304
 READTY 0144
 READ1 0063
 READ2 0104
 READ2A 0105
 RECORD 0040
 RECROK 0071
 REGENP 0064
 REPOSI 3400

MAINDEC-08-D3RA-LA

REREAD 0420
 REREDA 0470
 RESETD 3017
 RETJMP 3355
 RETUJM 0161
 REVCHK 2563
 REWCK 1517
 REWDRV 0525
 REWIND 0056
 ROT1TS 3222
 ROT2TS 3236
 RPATO 0700
 RSFDRV 3417
 RSFDR1 3435
 RSQFLG 0350
 SAMDRV 0325
 SAVNUM 1570
 SBRECD 3130
 SEARCH 3600
 SEKONS 4110
 SELGEN 1046
 SELRAN 0060
 SELVAR 1061
 SETIDX 3131
 SRCHER 0155
 SRCHIT 0055
 SRCHTY 0145
 SRCMOD 3755
 SRCONT 3663
 SREZTS 3726
 SRFBLK 3702
 SRHERR 4000
 SRTARN 3635
 SWCMDL 2223
 SWCMOD 2212
 TADTST 3252
 TAPONT 0156
 TEMP1 3341
 TEMP2 3342
 TEMP3 3343
 TEMP4 3344

TEMP5 3345
 TRECTR 2277
 TXSTOR 4320
 TYALL 4503
 TYBKW 4547
 TYCHAR 4230
 TYCONT 4272
 TYCOVR 4304
 TYDATA 4474
 TYDIR 4540
 TYDRV 4511
 TYFWD 4557
 TYMODE 4565
 TYMOVE 4400
 TYPCON 0106
 TYPTX 0107
 TYRALL 4432
 TYRDAT 4423
 TYREAD 4457
 TYSRCH 4411
 TYTEXT 4200
 TYWALL 4450
 TYWDAT 4441
 TYWRIT 4465
 UNFUNC 0046
 WAITI 0066
 WATINT 3000
 WCLOC 0043
 WDCERR 2253
 WDCMOD 2200
 WDINC 1503
 WRITES 0324
 WRITE1 1400
 WRTDTY 0101
 WRT1A 1455
 WSTERR 1526
 WTCNTR 3125
 WTKON1 3123
 WTKON2 3124
 XORSV 2565
 XORSUM 2531

11. FLOW CHARTS

